

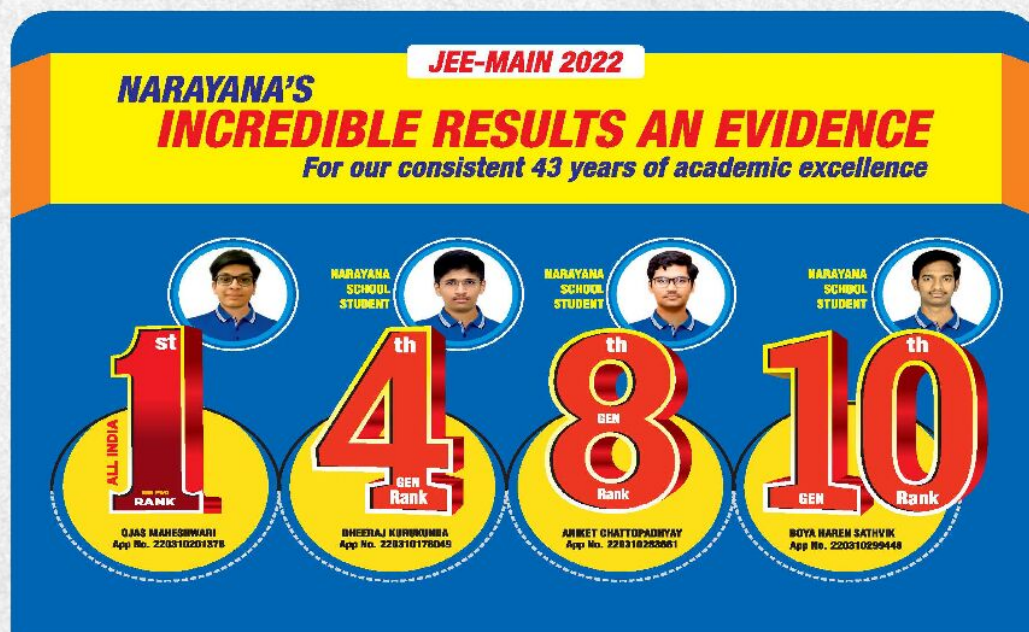


NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



ORGANIC CHEMISTRY PART-I (JR) ADVANCED MODEL QUESTIONS



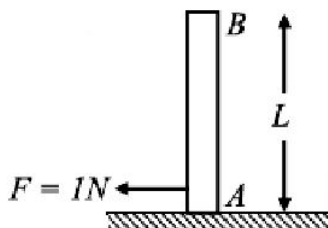
PHYSICS

HEAT, FLUIDS, ROTATIONAL DYNAMICS HEAT, FLUIDS, ROTATIONAL DYNAMICS
JEE ADVANCED IMPORTANT QUESTIONS

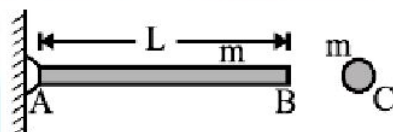
ROTATIONAL MOTION AND GRAVITATION

SINGLE ANSWER

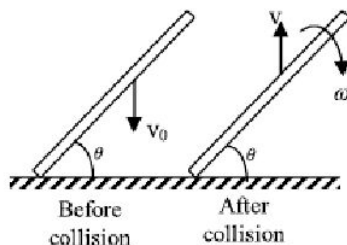
1. A uniform smooth rod of mass $m = 1\text{kg}$ and length L is balanced in the vertical position as shown in the figure when a horizontal force F is applied at end A , the acceleration of top point B at this instant is:



- A) $2m/s^2$ to right B) $1m/s^2$ to left C) $1m/s^2$ to right D) None of these
2. A uniform bar AB of mass m and a ball of the same mass are released from rest from the same horizontal position. The bar is hinged at end A. There is gravity downwards. What is the distance of the point from point B that has the same acceleration as the ball, immediately after release?



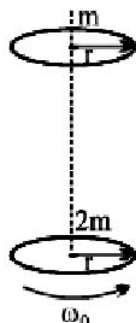
- A) $\frac{2L}{3}$ B) $\frac{L}{3}$ C) $\frac{L}{2}$ D) $\frac{3L}{2}$
3. A uniform bar of mass M and length L collides with a horizontal surface. Before collision velocity of centre of mass was v_0 and no angular velocity. Just after collision velocity of centre of mass of bar becomes v in upward direction as shown in the figure. Angular velocity ω of the bar just after impact is :



- A) $\frac{6(v_0 + v)\cos\theta}{L}$ B) $\frac{6(v_0 - v)\cos\theta}{L}$ C) $\frac{(v_0 + v)\cos\theta}{L}$ D) $\frac{(v_0 - v)\cos\theta}{6L}$

MULTIPLE

4. Which of the following statements is (are) true
- A) Work done by kinetic friction on a rigid body may be positive
- B) A uniform sphere rolls up an inclined plane without sliding. The friction force on it will be up the incline. (only contact force and gravitational force is acting)
- C) A uniform sphere rolls down an inclined plane without sliding. The friction force on it will be up the incline. (only contact force and gravitational force is acting)
- D) A uniform sphere is left from rest from the top of a rough inclined plane. It moves down the plane with slipping. The friction force on it will be up the incline always.
5. A disc of mass m and radius r is gently placed on another disc of mass $2m$ and radius r . The disc of mass $2m$ is rotating with angular velocity ω_0 initially. The disc is placed such that axis of both are coincident. The coefficient of friction is μ for surfaces in contact. Assume that pressure on disc is uniformly distributed find the correct statement.



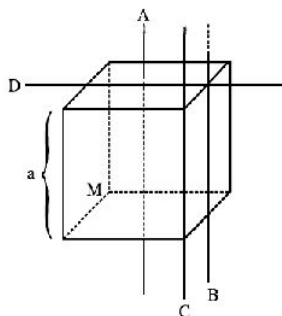
A) Loss in kinetic energy of system $\Delta K = \frac{1}{3}mr^2\omega_0^2$

B) Loss in kinetic energy of system $\Delta K = \frac{1}{6}mr^2\omega_0^2$

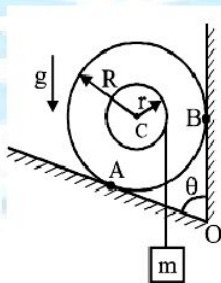
C) The common angular velocity is $\frac{2}{3}\omega_0$

D) The common angular velocity is $\frac{4}{3}\omega_0$

6. A uniform cubical block of mass M and side 'a' is as shown in the figure. Mark the correct statement (s)



- A) The moment of inertia about axis A, passing through the centre of mass is $I_A = \frac{1}{6}Ma^2$
- B) The moment of inertia about axis B, which bisects one of the cube faces is $I_B = \frac{5}{12}Ma^2$
- C) The moment of inertia about axis C, along one of the cube edges is $I_C = \frac{2}{3}Ma^2$
- D) The moment of inertia about axis D, which bisects one of the horizontal cube faces is $\frac{7}{12}Ma^2$
7. A massless spool of inner radius r , outer radius R is placed against vertical wall and tilted split floor as shown in the figure. A light inextensible thread is tightly wound around the spool through which a mass m is hanging. There exists no friction at point a, while the coefficient of friction between spool and point B is μ . The angle between two surface is θ .



A) the magnitude of force on the spool at B in order to maintain equilibrium is

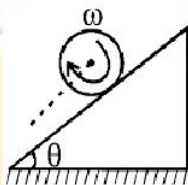
$$Mg\sqrt{\left(\frac{r}{R}\right)^2 + \left(1 - \frac{r}{R}\right)\frac{1}{\tan^2 \theta}}$$

B) the magnitude of force on the spool at B in order to maintain equilibrium is $mg\left(1 - \frac{r}{R}\right)\frac{1}{\tan \theta}$

C) the minimum value of μ for the system to remain in equilibrium is $\frac{\cot \theta}{(R/r) - 1}$

D) the minimum value of μ for the system to remain in equilibrium is $\frac{\tan \theta}{(R/r) - 1}$

8. A solid sphere is given an angular velocity ω and kept on a rough fixed incline plane as shown in the figure. Then choose the correct statement.



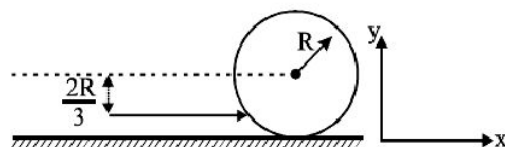
A) If $\mu = \tan \theta$ then sphere will be in linear equilibrium for some time and after that pure rolling down the plane will start.

B) If $\mu = \tan \theta$ then sphere will move up the plane and frictional force acting all the time will be $2mg \sin \theta$.

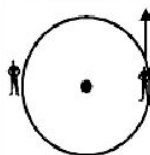
C) If $\mu = \frac{\tan \theta}{2}$ there will never be pure rolling (consider inclined plane to be long enough.)

D) If incline plane is not fixed and it is on smooth horizontal surface then linear momentum of the system (wedge and sphere) can be conserved in horizontal direction.

9. A billiard ball initially at rest is given a sharp blow by a cue stick. The force is horizontal and is applied at a distance $\frac{2R}{3}$ below the centre line, as shown in the figure. The initial speed of the ball is v_0 , and the coefficient of kinetic friction is μ_k . (μ_k is small):

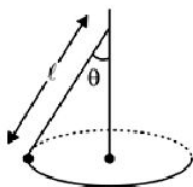


- A) Initially kinetic friction acts in $-\hat{i}$ direction B) Initially kinetic friction in $\hat{i}$ direction.
 C) Ball instantaneously starts pure rolling D) Initial angular velocity of ball is $\frac{5v_0}{3R}$
10. A student Amir of mass m is standing on the edge of an horizontal disc of radius R . The disc is free to rotate about a frictionless vertical axis passing through its centre. Initially the student and disc are at rest. At $t = 0$ the student starts to run at his maximum velocity v relative to disc. He runs along the circumference of the disc towards his friend Salman in anticlockwise sense. Salman is stationary on ground. Moment of inertia of disc is I .

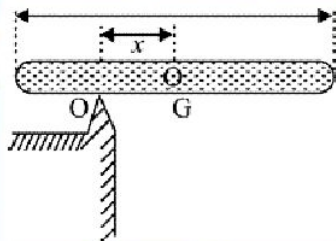


- A) Time taken to reach Salman is $\frac{\pi R}{v} \left(1 + \frac{I}{mR^2} \right)$.
 B) If disc were considered massless time taken to reach Salman is $\frac{\pi R}{v}$.
 C) If disc were considered massless Amir can never reach Salman.
 D) Time taken will be minimum if Amir instead of running along circumference runs along diameter towards Salman.

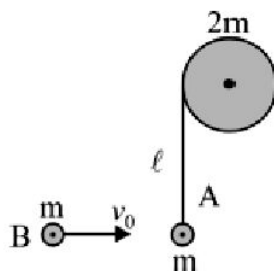
11. A particle is rotating in a conical pendulum with help of a string of length l as shown in the figure. The speed of the particle is constant and angle θ is also constant with time. It can be said that



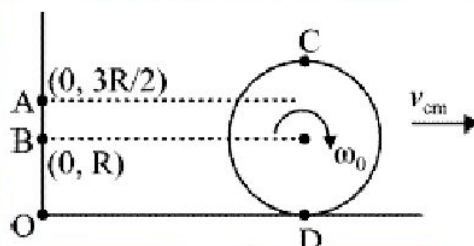
- A) Angular momentum of the particle about the point of suspension is not constant.
 B) Only the direction of angular momentum of the particle about the point of suspension is constant.
 C) Only the magnitude of angular momentum about the point of suspension is constant.
 D) Net torque on the particle about the point of suspension is zero.
12. In the adjacent figure a uniform rod of length l and mass m is kept at rest in horizontal position on an elevated edge. The value of x (consider the figure) is such that the rod will have maximum angular acceleration α , as soon as it is set free. (friction is sufficient)



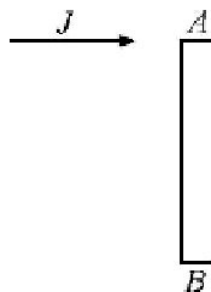
- A) x is equal to $\frac{l}{2\sqrt{3}}$ B) α is equal to $\frac{g\sqrt{3}}{2l}$
 C) α is equal to $\frac{g\sqrt{3}}{l}$ D) x is equal to $\frac{l}{\sqrt{3}}$
13. In the adjacent figure, a uniform disc of mass $2m$ and radius $\frac{l}{2}$ is lying at rest on a smooth horizontal surface. A particle 'A' of mass m is connected to a light string of length l , whose other end is attached to the circumference of the disc. Initially string is just taut and tangential to the disc, particle A is at rest. In the same horizontal plane another particle B of same mass m moving with velocity v_0 perpendicular to string collides elastically with A. Just after impact which of the following statements will be true



- A) Tension in the string is $\frac{2mv_0^2}{5l}$
- B) Acceleration of the centre of the disc is $\frac{v_0^2}{5l}$
- C) Tension in the string is $\frac{mv_0^2}{5l}$
- D) Acceleration of the centre of the disc is $\frac{2v_0^2}{5l}$
14. A cylinder is rolling without sliding on a smooth horizontal surface as shown. O is origin, B and A are two points on y – axis C and D are topmost and bottom most point of cylinder at the given instant. Mark the correct statements.

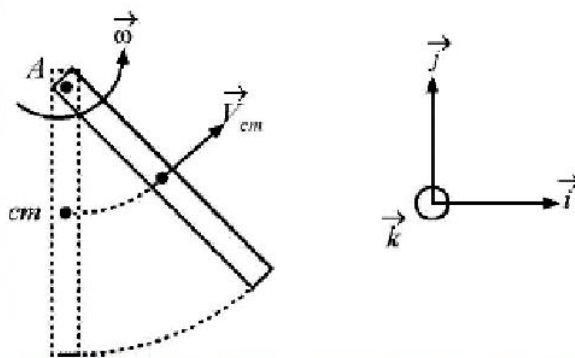


- A) Angular momentum of body about C and D will be same in magnitude
- B) Angular momentum of cylinder will be minimum about A among the given points
- C) Angular momentum will be minimum about B
- D) Angular momentum about O and C will be same in magnitude
15. A rod AB of mass M and length L lies on a smooth horizontal table. An impulse J is applied to end A as shown in the figure immediately after imparting the impulse:



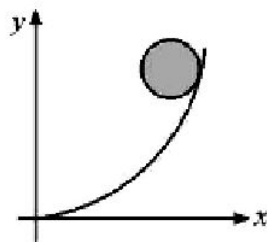
- A) The radius of curvature of trajectory of A as seen from the ground is $\frac{8}{9}L$.
- B) The radius of curvature of trajectory of B as seen from the ground is $\frac{2}{9}L$.
- C) The instantaneous axis of rotation is at a distance of $\frac{L}{6}$ from the mid-point of the rod.
- D) The mid-point of the rod will move along a straight line.

16. A uniform rod of length l and mass m is rotating about a fixed axis perpendicular to the paper (xy-plane) and passing through its one of the end A . At the instant shown in the figure rod's angular velocity is $\vec{\omega}$ and velocity of the centre of mass of the rod is $\vec{v}_{cm}$. The moment of inertia of the rod about an axis passing through point A and kinetic energy of rod respectively.



- A) $\vec{L}_A = I_A \vec{\omega} = \left(I_{cm} + \frac{ml^2}{4} \right) \vec{\omega}, K = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$
- B) $\vec{L}_A = I_A \vec{\omega} = m \vec{r}_{cm} \times \vec{v}_{cm} + I_{cm} \vec{\omega}, K = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$
- C) $\vec{L}_A = I_A \vec{\omega} = \left(I_{cm} + \frac{ml^2}{4} \right) \vec{\omega}, K = \frac{1}{2} I_A \omega^2$
- D) $\vec{L}_A = I_A \vec{\omega} = m \vec{r}_{cm} \times \vec{v}_{cm} + I_{cm} \vec{\omega}, K = \frac{1}{2} I_A \omega^2$

17. A small disc of mass m is released on a parabolic curve in a vertical plane such that gravity acts along negative y - axis. The equation of parabolic curve is $x^2 = \frac{2a}{\sqrt{3}} y$, where ' a ' is a positive constant. Frictional force between disc and curve are sufficient for pure rolling. When disc is reached at $x=a$ then choose the correct option (s)



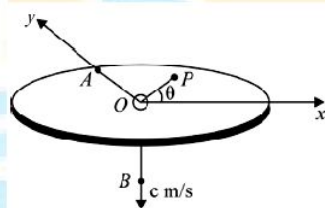
A) Acceleration of disc along the trajectory is $\sqrt{3}g$

B) Acceleration of disc along the trajectory is $\frac{g}{\sqrt{3}}$

C) Frictional force between disc and curve is $\frac{mg}{2\sqrt{3}}$

D) Frictional force between disc and curve is $\frac{mg}{\sqrt{3}}$

18. A small object is placed on a smooth table at a distance r_0 from a hole in the centre of table. An ideal inextensible chord is attached to the object through this hole. The object is set into motion with initial velocity of magnitude v_0 at right angle to the chord and at the same time, the chord is pulled through the hole at uniform speed c m/s. Initially object is at point $A(0, r_0)$ and at any time t , it is at point $P(r \cos \theta, r \sin \theta)$. Neglect the dimension of object. Choose CORRECT option (s)



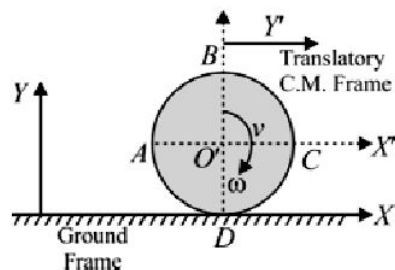
A) Tension in the chord at any time t is $\frac{mr_0^2 v_0^2}{(r_0 - ct)^3}$

B) Tension in the chord at any time t is $\frac{2mr_0^2 v_0^2}{(r_0 - ct)^3}$

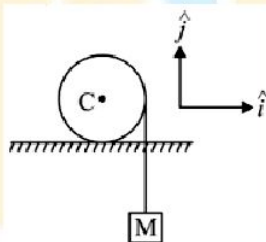
C) Angular speed of the object at any time t is $\frac{2r_0 v_0}{(r_0 - ct)^2}$

D) Angular speed of the object at any time t is $\frac{r_0 v_0}{(r_0 - ct)^2}$

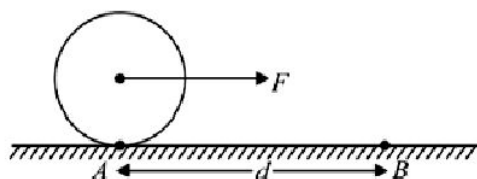
19. A thin rigid uniform circular disc rolls without slipping on a horizontal rigid surface (or the ground). At a certain instant, its position w.r.t. ground frame is as shown in the figure.



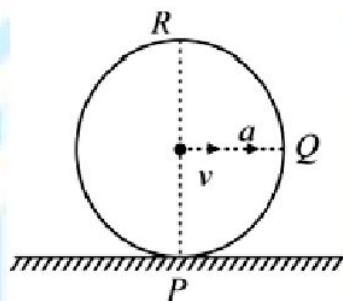
- A) Sector ABC has greater kinetic energy than sector ADC w.r.t. ground frame
 B) Sector BO'C has greater kinetic energy than sector CO'D w.r.t. ground frame
 C) Sector BO'C has the same kinetic energy as sector AO'B w.r.t. ground frame
 D) All the sectors AO'B, BO'C, CO'D and AO'D have same kinetic energy w.r.t. the centre of mass frame
20. A disc of mass M and radius r has massless string wrapped over it with one end fixed on disc and other end connected to block of same mass M. Initially system is held at rest. Now system is released from rest. Immediately after system is released (assume there is no slipping at any contact surface).



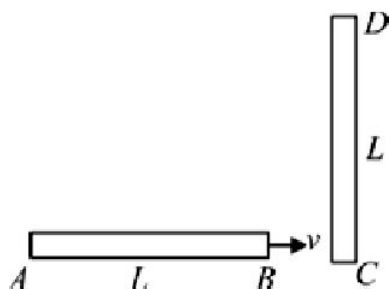
- A) Acceleration of block in ground frame is $\frac{2}{5}g\hat{i} - \frac{2}{5}g\hat{j}$
 B) Acceleration of block in the frame of centre of disc is $-\frac{2}{5}g\hat{i} - \frac{2}{5}g\hat{j}$
 C) Acceleration of block in ground frame is $-\frac{2}{5}g\hat{j}$
 D) Tension in string is $3mg/5$
21. A uniform ring is connected to a light axle with light spokes so as to form a wheel. The wheel is placed on a horizontal surface with its plane vertical and a constant horizontal force F is applied to the axle. Surface AB is rough and surface right to B is smooth. The wheel does not slip when it moves from A to B and takes time T to move the distance d between A and B.



- A) Rotational KE remains constant right of B
 B) Energy is dissipated from A to B
 C) Linear acceleration of centre of mass of the ring right of B is more than linear acceleration left of B
 D) Linear acceleration of centre of mass of the wheel undergoes constant non-zero acceleration to the right of B.
22. A uniform wheel is rolling without slipping on a horizontal surface. At a certain instant, its centre of mass has velocity ' v ' and acceleration ' a '. P , Q and R are three points on the rim of the disc as shown in the figure. Acceleration of



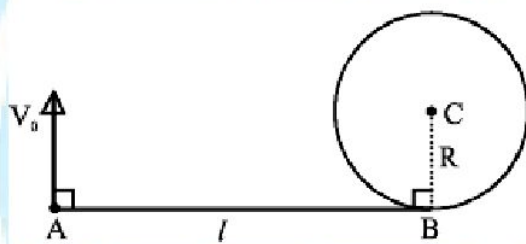
- A) P is vertically upwards
 B) Q may be vertically downwards
 C) R cannot be horizontal
 D) Some point on the rim may be horizontal leftwards.
23. A rod CD of length L and mass M is placed horizontally on a frictionless horizontal surface as shown. A second identical rod AB which is also placed horizontally (perpendicular to CD) on the same horizontal surface is moving along the surface with a velocity v in a direction perpendicular to rod CD and its end B strikes the CD at end C and sticks to it rigidly. Then,



- A) Velocity of centre of mass of the system just after impact is $\frac{v}{4}$.
- B) The ω (angular speed) of system just after collision is $\frac{3v}{5L}$.
- C) Velocity of centre of mass of the system just after impact is $\frac{v}{2}$.
- D) The ω (angular speed) of system just after collision is $\frac{5v}{3L}$.

PARAGRAPH : 24 – 26

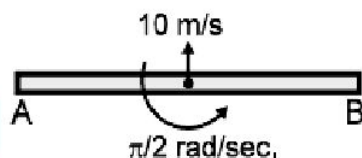
A particle A is attached through an ideal thread to a point B on the circumference of a smooth vertical cylinder as shown in the figure. Thread is horizontal and the particle can slide on a smooth horizontal plane. The particle is now given a horizontal speed V_0 perpendicular to thread as shown. For the subsequent motion, answer the following questions: (l = length of thread, R = radius of cylinder):



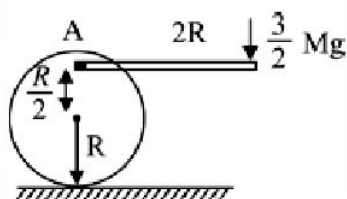
24. Angular momentum of particle is conserved about point :
 A) B B) C C) Midpoint of BC D) None of these
25. From start upto the moment particle strikes the cylinder, external torque required to keep the cylinder in equilibrium :
 A) It always zero B) Continuously increase
 C) First increases then decreases D) Remains constant
26. If the motion starts at $t = 0$, the time at which particle strikes the cylinder is :
 A) $\frac{l^2}{RV_0}$ B) $\frac{l^2}{2RV_0}$ C) $\frac{l^2}{3RV_0}$ D) None of these

NUMERICAL

27. A uniform rod AB of length 4m and mass 12 kg is thrown such that just after the projection the centre of mass of the rod moves vertically upwards with a velocity 10 m/s and at the same time it is rotating with an angular velocity $\frac{\pi}{2}$ rad/sec about a horizontal axis passing through its midpoint. Just after the rod is thrown it is horizontal and is as shown in the figure. Find the acceleration (in m/sec^2) of the point A in m/sec^2 when the centre of mass is at the highest point. (Take $g=10\text{m/sec}^2$ and $\pi^2 = 10$)



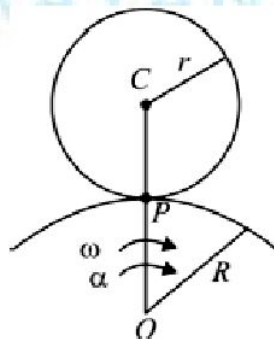
28. A disc of mass M and radius R is placed on a rough horizontal surface. A light rod of length $2R$ is fixed to the disc at point A as shown in the figure and force $\frac{3}{2}Mg$ is applied at the other end. Find the minimum value of 10μ where μ is coefficient of friction (upto one decimal place) between disc and horizontal surface so that disc starts to roll without slipping.



29. A uniform cylinder of mass m and radius r rolls on a fixed cylindrical surface of radius R . at a certain instant, the line OC has an angular velocity ω and angular acceleration α . Find :

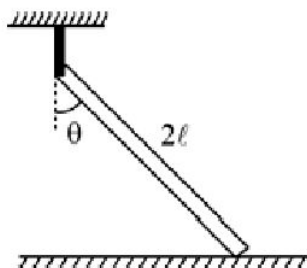
If the acceleration of point of contact P of the cylinder with respect to the surface is $\frac{7R(R+r)\omega^2}{kr}$

Then find value of k .

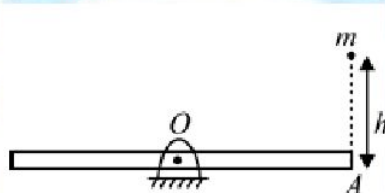


30. A uniform rod of length $2l$ and mass m is suspended from one end by inextensible string and other end lies on smooth ground. The angle made by rod with vertical is $\theta = \sin^{-1}(1/\sqrt{3})$. If N_1 and N_2

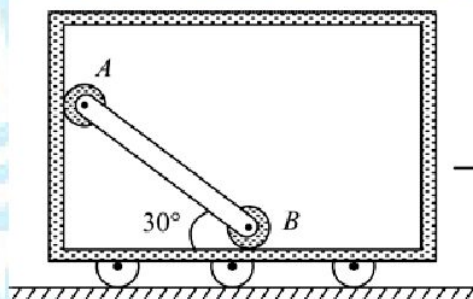
represents the contact force from ground on rod just before and just after cutting string then find the ratio of N_1 / N_2 .



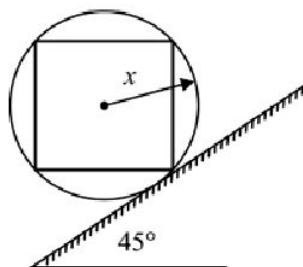
31. A uniform rod of length 12 metre and mass m is hinged at midpoint and lies horizontally at rest. Rod is free to rotate in vertical plane about hinge O . A particle of mass m is released from height h as shown in figure. Particle collides with end A and sticks to it. Calculate minimum height h (in metre) so that system can complete vertical circular motion.



32. If a car is moving rightward with acceleration $a = g\sqrt{k}$ m/s² rightward as shown in the figure. Find the value of k so that, rod maintains its orientation as shown in the figure. Neglect the friction and mass of the small roller at A and B .



33. For identical rods, each of mass m are welded at their ends to form a square, and the corners are then welded to a light metal hoop of radius r . If the rigid assembly of rods and hoop is allowed to roll down the inclined rough surface. If the minimum value of the coefficient of static friction which will prevent slipping is $\frac{k}{10}$. Find the value of k .



ROTATIONAL MOTION AND GRAVITATION

KEY

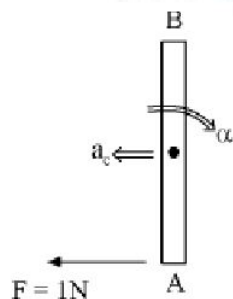
1	2	3	4	5	6	7	8	9	10
A	B	A	ABCD	BC	ABC	AC	AD	AD	CD
11	12	13	14	15	16	17	18	19	20
AC	AC	AB	ABD	ABCD	ABCD	BC	AD	ABCD	BCD
21	22	23	24	25	26	27	28	29	30
ACD	ABCD	BC	D	B	B	5	8	7	1
31	32	33							
8	3	4							

HINTS

$$1. \quad F \cdot \frac{l}{2} = \frac{ml^2}{12} \alpha \Rightarrow \alpha = \frac{6F}{ml}$$

$$F = ma_c \Rightarrow a_c = \frac{F}{m}$$

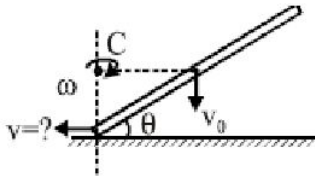
$$a_B = \frac{la}{2} - a_c = \frac{3F}{m} - \frac{F}{m} = 2m/s^2 \text{ (right)}$$



$$2. \quad mg \frac{L}{2} = \frac{1}{3} mL^2 \alpha$$

$$\therefore \alpha = \frac{3g}{2L} \text{ and } \alpha x = g \Rightarrow \frac{3g}{2L} x = g \therefore x = \frac{2L}{3}$$

$$\therefore \text{Distance from } B = \frac{L}{3}$$

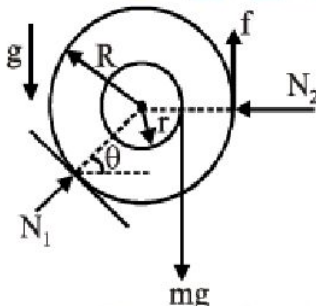


3. Impulse = $mV - (-mV_0) = m(V + V_0)$ and about centre of mass angular impulse

$$= m(V + V_0) \frac{l}{2} \cos \theta = \frac{mV^2}{12} \cdot \omega \quad (= \text{change in angular momentum}) \Rightarrow \omega = \frac{6(V + V_0) \cos \theta}{l}$$

4. Work done by kinetic friction on ONE body may be positive / negative / zero. Direction of frictional force in B,C,D is correct for providing the necessary torque.
 5. The angular momentum of the system is conserved. Kinetic energy will not be conserved because friction is there.
 6. Parallel Axis theorem, check the distance carefully. $I_D = I_B$ (symmetric)

7. $mgr = fR$ (i)
 $N_1 \sin \theta + f = mg$ (ii)
 $N_1 \cos \theta = N_2$ (iii)
 $f \leq \mu N_2$ (iv)



8. Due to torque of friction about CM ω eventually decreases to zero, initially there is no translation. Friction is sufficient for pure rolling therefore after sometime pure rolling begins. There is no external force in $\times$ direction therefore momentum is conserved along $\times$ direction.
 9. Angular impulse = $(2R/3)$ Linear impulse. This will give the angular speed of sphere just after collision. Impulse of friction DURING collision is negligible.

10. Conceptual

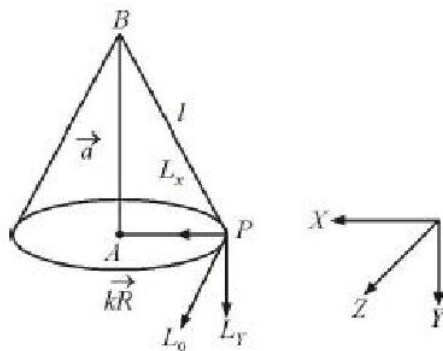
11. AB is $\vec{a}$

$$AP \text{ is } \vec{R}$$

At point P, velocity is out of the plane

$$\text{Angular momentum } \vec{L}_0 = \vec{L} \times \vec{mv} = (\vec{a} + \vec{R}) \times \vec{mv}$$

$$\vec{L}_0 = \underbrace{(\vec{R} \times \vec{m}\vec{v})}_{\text{Constant in direction on magnitude}} + \underbrace{(\vec{a} \times \vec{m}\vec{v})}_{\text{Constant in magnitude only}}$$



12. $I_0 \alpha = \tau_0$

$$\left[\frac{ml^2}{12} + mx^2 \right] \alpha = mgx \dots (i)$$

$$\alpha = \frac{12gx}{l^2 + 12x^2}$$

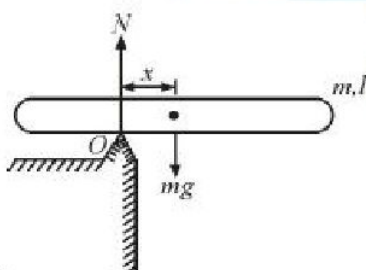
For α maximum

$$\frac{d\alpha}{dt} = 0 ; \quad \frac{d}{dx} \left[\frac{12gx}{l^2 + 12x^2} \right] = 0$$

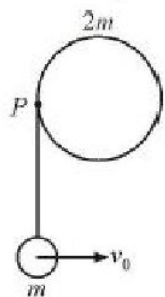
$$\frac{12}{12x^2 + l^2} - \frac{288gx^2}{(12x^2 + l^2)^2}$$

$$\frac{12g(12x^2 - l^2)}{(12x^2 + l^2)^2} = 0 ; 12x^2 - l^2 = 0 ; x = \frac{l}{2\sqrt{3}}$$

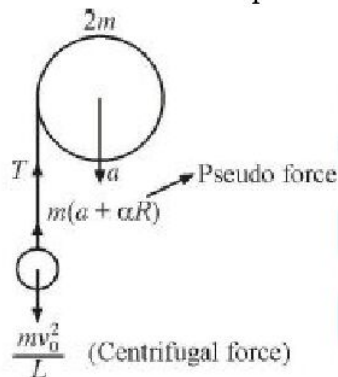
Now put x in (i) ; $\alpha = \frac{g\sqrt{3}}{l}$



13. Velocity of connected mass will be v_0 due to conservation of linear momentum



Drawn FBD with respect to P.



(i) $TR = I\alpha$ For $2m$ (ii) $T = 2ma$

(ii) $T + m(a + dR) = \frac{mv_0^2}{L}$ for $m \Rightarrow T = \frac{2mv_0^2}{5l}$ and $a = \frac{v_0^2}{5l}$

Also after certain value of F, torque on cylinder will be constant hence D is correct.

14. $\omega_0 R = v_{com}$

$$L = mv \times R + I\omega$$

$$L_C = mv_{com}R(-\hat{k}) + I_{com}\omega_0(-\hat{k})$$

$$L_D = mv_{com}R(-\hat{k}) + I_{com}\omega_0(-\hat{k})$$

$$L_C = L_D$$

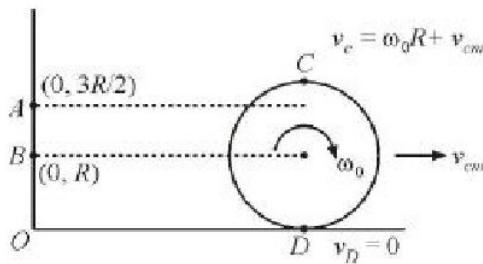
$$L_0 = +mv_{com}R(-\hat{k}) + I_{com}\omega_0(-\hat{k})$$

$$L_A = +mv_{com}R(-\hat{k}) + I_{com}\omega_0(-\hat{k})$$

$$L_B = I_{com}\omega_0(-\hat{k})$$

So L_A is minimum

$$L_0 = L_C$$



15. $R = \frac{v^2}{a_c}$; $v = \frac{J}{M}$

$$\omega = \frac{\frac{J}{M}}{\frac{2}{12}} = \frac{6J}{ML} \Rightarrow v_A = v + \frac{\omega L}{2} = \frac{J}{M} + \frac{3J}{M} = \frac{4J}{M} ; a_{c_A} = \frac{\omega^2 L}{2} = \frac{18J^2}{M^2 L}$$

$$\Rightarrow R_A = \frac{v_A^2}{a_{c_A}} = 16 \frac{J^2}{M^2} \cdot \frac{M^3 L}{18J^2} = \frac{8}{9} L ; v_B = v - \frac{\omega L}{2} = \frac{2J}{M} \Rightarrow R_B = \frac{4J^2}{M^2} \cdot \frac{M^3 L}{18J^2} = \frac{2}{9} L$$

16. Basic knowledge to write angular momentum and kinetic energy of the system.

17. $x^2 = \frac{2a}{\sqrt{3}} y$; $2x = \frac{2}{\sqrt{3}} a \frac{dy}{dx}$; $\tan \theta = \sqrt{3} \frac{x}{a}$

$$\theta = 60^\circ ; a = \frac{g \sin \theta}{1 + \frac{I_{cm}}{mR^2}} = \frac{g}{\sqrt{3}} ; f = \frac{mg \sin \theta}{1 + \frac{mR^2}{I_{cm}}} = \frac{mg}{2\sqrt{3}}$$

18. $r = r_0 - ct \dots (i)$ ($\because$ distance covered by thread in time t)

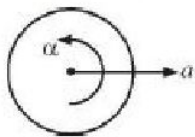
By conservation of angular momentum about O ,

$$I_0 \omega_0 = I \omega$$

$$\left(m r_0^2 \right) \left(\frac{v_0}{r_0} \right) = m r^2 \left(\frac{v}{r} \right)$$

$$v_0 r_0 = v r \Rightarrow v = \frac{v_0 r_0}{r} \dots (ii)$$

T at any time $= m v^2$ velocity and radius, r at that instant



$$T \text{ and time } T = \frac{m v^2}{r} = \frac{m \left(\frac{v_0 r_0}{r} \right)^2}{r} = \frac{m v_0^2 r_0^2}{r^3}$$

$$T = \frac{m v_0^2 r_0^2}{(r_0 - ct)^3} \dots (ii)$$

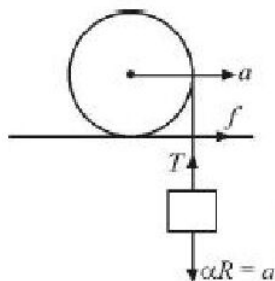
$$\omega \text{ at time } \omega = \frac{v}{r} = \frac{v_0 r_0}{r^2}$$

$$\omega = \frac{v_0 r_0}{(r_0 - ct)^2} \quad [\text{From (i) and (ii)}]$$

$$19. \sqrt{v_{com}^2 + (\omega R)^2 + 2v_{com}\omega R \cos \theta}$$

In frame of ωR com each point has velocity and hence same KE.

$$20. a = \alpha R \quad (\because \text{no slipping})$$



$$\text{For block, } ma = mg - T \quad \dots (i)$$

$$\text{For disc, } TR - Fr = I\alpha$$

$$T - f = \frac{mR^2}{2} \times \frac{a}{R} \times \frac{1}{R}$$

$$T - f = \frac{ma}{2} \quad \dots (ii)$$

$$\text{And } f = ma \quad \dots (iii)$$

$$\text{Put (i) and (iii) in (ii) ; } a = \frac{2}{5}g$$

$$\text{For block, acceleration } \rightarrow -\frac{2}{5}g \hat{j}$$

$$\text{For disc, acceleration } \rightarrow \frac{2}{5}g \hat{j}$$

Acceleration of block in frame of disc

$$a_{BD} = a_B + a_D$$

$$= -\frac{2}{5}g \hat{j} - \frac{2}{5}g \hat{i}$$

$$\text{From (i) } T = mg - ma$$

$$T = mg - \frac{2mg}{5} ; T = \frac{3mg}{5}$$

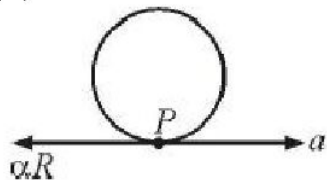
$$21. \text{ To the right } B, \text{ there is no friction therefore, no torque acts on the body}$$

$\therefore$ The angular velocity remains constant (no angular acceleration)

Rotation KE will be constant to the right of B

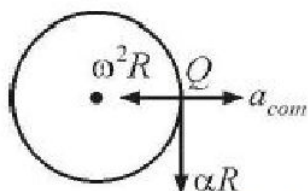
To the right B , F is still being applied; therefore the object will still undergo constant linear acceleration.

22. (A) is correct since $v = \omega R$ at P any horizontal acceleration at P will cause slipping $a = \alpha R$

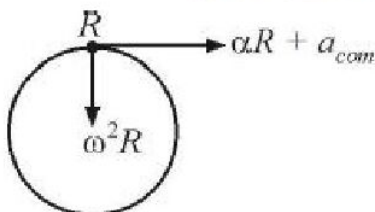


- (B) If $\omega R^2 = a_{com}$

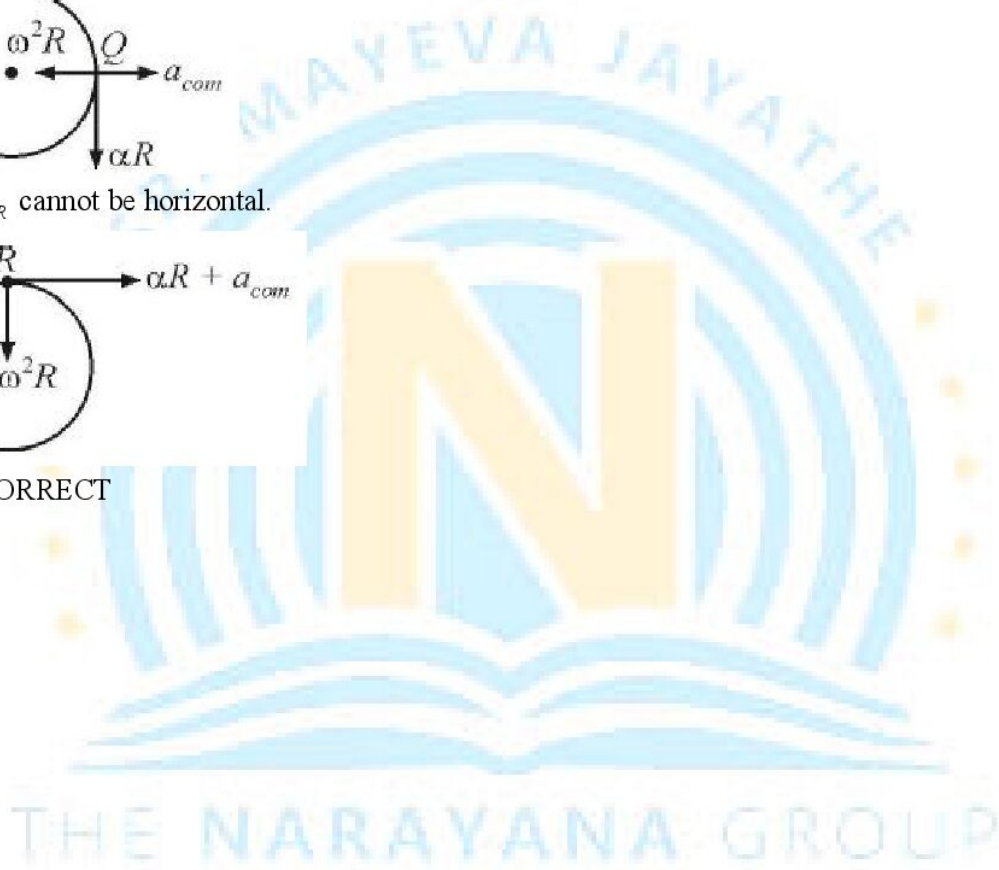
a_Q will be downward



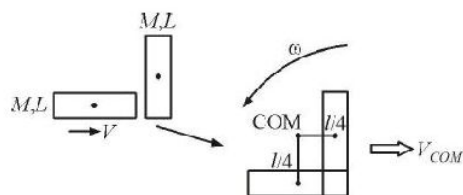
- (C) a_R cannot be horizontal.



- (D) CORRECT



23.



Conservation linear momentum

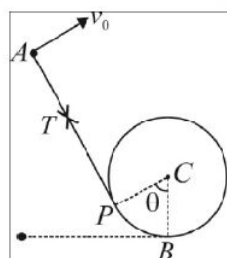
$$MV = 2MV_{COM}$$

$$V_{COM} = \frac{V}{2}$$

Conservation of angular momentum

$$MV_{l/4} = 2 \left[\frac{ML^2}{12} + \frac{ML^2}{8} \right] \omega ; \quad \omega = \frac{3V}{5L}$$

24 - 26



Let us first understand some general concepts of the problem.

Speed of the particle remains constant. Since the only force acting on the particle is tension and this force is always perpendicular to the instantaneous velocity of the particle. Hence tension does not work on the particle and by work energy theorem; speed of the particle remains constant.

Let us denote the point at which the tread touches the cylinder by P. As we can see, the speed as well as acceleration of this point is zero. Hence, at an instant, in the reference frame of this point, the particle can be taken to be performing circular motion

(A) Torque on the particle is due to T and obviously NOT zero about B, C, or midpoint of BC. Hence answer to 'a' is none of these.

$$(B) \quad T = \frac{mv_0^2}{r} = (r = \text{length } AP).$$

$\therefore r$ continuously decrease whereas m, v_0 remain constant, T continuously increases. Torque on cylinder due to T is TR. So, this torque also continuously increases. Hence, the external torque required to keep the cylinder stationary (by balancing the torque TR) should also be increased continuously.

$$(C) \quad \text{At any instant, angular speed of segment AP is: } \omega = \frac{v_0}{r} \text{ where } r + R\theta = l \Rightarrow r = (l - R\theta)$$

$$\text{So, } \frac{d\theta}{dt} = \frac{v_0}{l - R\theta} \dots \dots (i)$$

$$\theta \text{ goes from zero to } \theta = \frac{l}{R}. \therefore \int_0^{l/R} (l - R\theta) d\theta = v_0 \int_0^T dt \quad \text{We can get } \boxed{T = \frac{l^2}{2Rv_0}}$$

Note: $\because$ PA is always perpendicular to PC, angular speed of PC and PA are same. That is why θ on both sides of equation (i) are taken to be same.

27. Time in which C.M. reaches it highest point = 1 sec. (from $v = u + at$, putting $v = 0$, $u = +10$, $a = 10 \text{ m/sec}^2$) after projection angular velocity will not change as the torque of external forces is zero.

In 1 sec., the rod will rotate by an angle $= \omega t = \frac{\pi}{2} \times 1 \text{ rad.}$

The rod will be vertical with point A at the lowest point.

$$\therefore a_A = g - \omega^2 L/2 = 10 - \frac{\pi^2 4}{4 \times 2} = 5 \text{ m/sec}^2$$

28. For point O $\frac{3mg}{2} + mg = N$

$$N = \frac{5mg}{2} ; f = \frac{5}{2} \mu gm$$

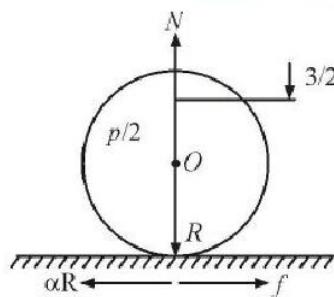
$$\alpha R = a_{com} ; a_{com} = \frac{f}{m}$$

$$I\alpha = \frac{3}{2} mg \times 2R - fR$$

$$\frac{mR^2}{2} \alpha = \frac{3}{2} mg \times 2R - \frac{5}{2} \mu mg R ; \alpha = \frac{6g - 5\mu g}{R}$$

$$\alpha R = \frac{f}{m} = 6g - 5\mu g ; 6mg - 5\mu g = \frac{5}{2} \mu mg$$

$$\mu = \frac{12}{15} = \frac{4}{5} ; 10\mu = 8$$



29. $\frac{R(R+r)\omega^2}{r} ; V_C = \omega(R+r)$

Also, $V_C = \omega' r$

Where ω' is the angular velocity of cylinder about C

$$\Rightarrow \omega' r = \omega(R+r) \Rightarrow \omega' = \frac{\omega(R+r)}{r}$$

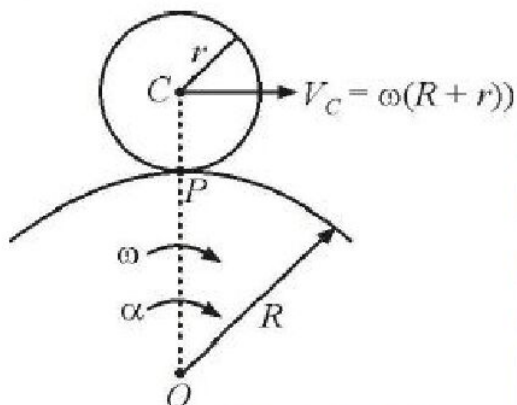
a_x of point $P = 0$ (pure rolling) acceleration of P along Y w.r.t. C is $\omega^2 r$ vertical upward.

However C itself has acceleration in $Y = \omega^2 (R + r)$ vertically downward

$$(a_p / c) \gamma + \vec{a}_C = a_p$$

$$\omega^2 r - \omega^2 (R + r) = a_p$$

$$\left\{ \frac{\omega(R+r)}{r} \right\}^2 r - \omega^2 (R+r) = a_p \Rightarrow a_p = \frac{R(R+r)\omega^2}{r} ; \frac{7R(R+r)\omega^2}{kr}$$



30. Before cutting, apply equation of translational and rotational equilibrium about point P.

$$N_1 + T = mg \text{ and } T(2l)\sin\theta = mgl\sin\theta$$

$$\Rightarrow N_1 = mg/2 \dots\dots(i)$$

After cutting, the acceleration of centre of mass will be vertically downwards.

$$Z \text{ about } POC = T\alpha$$

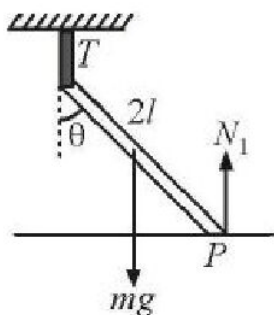
$$\frac{ml^2}{3} \alpha \dots(ii)$$

$$\text{And } a_{com} = \frac{\alpha l}{2} \dots\dots(iii)$$

Now, along vertical

$$ma_{com} = mg - N_2 \dots(iv)$$

Solve equation (i), (ii), (iii) and (iv)



31. For critical case

Initial KE=P.E. at max height

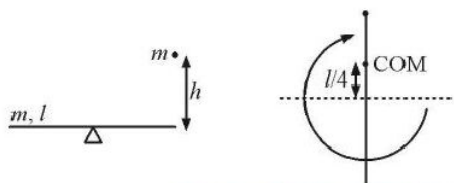
$$mg \frac{l}{4} = \frac{1}{2} I_{Hinge} \omega^2$$

Angular momentum conservation about hinge

$$mv \frac{l}{2} = I \omega$$

$$m\sqrt{2gh} \frac{l}{2} = \left[\frac{ml^2}{12} + \frac{ml^2}{4} \right] \omega \quad (v = \sqrt{2gh} \because v^2 - u^2 = 2gh)$$

$$\frac{6}{4l} \sqrt{2gh} = \omega ; 2mg \frac{l}{4} = \frac{1}{2} \left[\frac{ml^2}{12} + \frac{ml^2}{4} \right] \left[\frac{36}{16l^2} \times 2gh \right] ; \frac{2l}{3} = h = \frac{2}{3} \times 12 = 8 ; h = 8$$



32. Length = $2l$

Apply pseudo force ma to left at centre of mass of rod by translational equilibrium,

$$N_1 = ma \quad \dots(i)$$

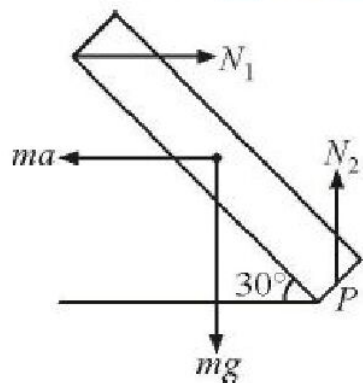
By rotational equilibrium about point P,

$$\because mal \sin 30^\circ + mgl \cos 30^\circ = N_1 (2l) \sin 30^\circ$$

$$\frac{mal}{2} + \frac{mgl\sqrt{3}}{2} = N_1 l \quad \dots(ii)$$

Put (i) and (ii)

$$a = g\sqrt{3} \Rightarrow R = 3$$



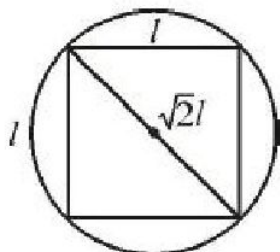
33. Moment of inertia each rod

$$= \frac{ml^2}{12} + m \left(\frac{l}{2} \right)^2$$

$$= \frac{2mr^2}{3} \quad (\because l = \sqrt{2}r)$$

For entire object,

$$I = 4 \times \left(\frac{2mr^2}{3} \right) = \frac{8mr^2}{3}$$



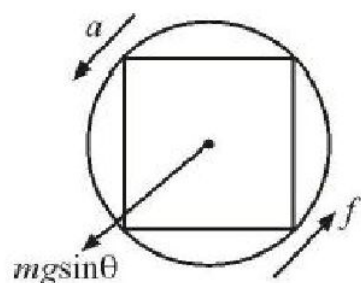
$$\sqrt{2}l = 2r$$

$$\Rightarrow l = \sqrt{2}r$$

Now, $4mg \sin \theta - f \dots (i)$

And $fR = I\alpha \Rightarrow f = \frac{Ia}{R^2}$

$$\Rightarrow f = \frac{8ma}{3} \dots (ii)$$



Putting (ii) in (i), $\frac{20ma}{3} = 4mg \sin \theta \Rightarrow a = \frac{12g}{20\sqrt{2}}$

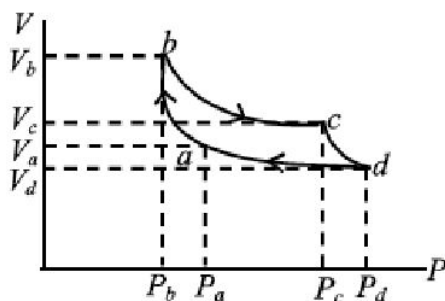
Putting in (2), $f = \frac{8m}{3} \times \frac{12g}{20\sqrt{2}} = \frac{8mg}{5\sqrt{2}}$

Now, $f \leq \mu N \Rightarrow f \leq \mu mg \cos \theta \times 4 \Rightarrow \frac{8mg}{5\sqrt{2}} \leq \frac{\mu mg}{\sqrt{2}} \times 4 \Rightarrow \mu = \frac{4}{10}$

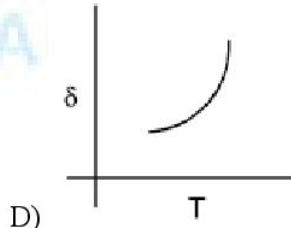
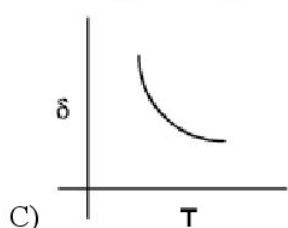
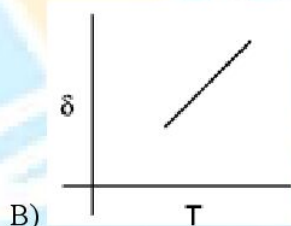
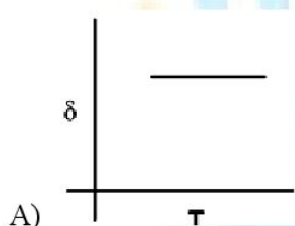
GASEOUS STATE & THERMODYNAMICS

SINGLE CORRECT

1. A cyclic process $ABCD$ shown in $V-P$ diagram with two adiabatic and two isothermal process for n mole of monoatomic gas is shown in the figure. The ratio of $\frac{P_b}{P_c}$ is :



- A) $\frac{P_a}{P_d}$ B) $\left(\frac{P_a}{P_d}\right)^{\gamma-1}$ C) $\frac{P_d}{P_a}$ D) $\left(\frac{P_d}{P_a}\right)^{\gamma}$
2. An ideal gas is initially at a temperature T and volume V . Its volume is increased by ΔV due to an increase in temperature ΔT , pressure remains constant. The quantity $\delta = \frac{\Delta V}{V\Delta T}$ varies with temperature as:



3. The average degree of freedom per molecule of a gas is 6. The gas performs 25 J of work at constant pressure. The heat absorbed by the gas is
- A) 75 J B) 100 J C) 150 J D) 125 J
4. An ideal gas has molar heat capacity C_v at constant volume. The gas undergoes the process $T = T_0(1 - \alpha V)$ where α , T_0 is constant and V is volume. Then molar heat capacity of the gas is:

A) $C_v + \frac{R}{\alpha V}(1 - \alpha V)$

B) $C_v - \frac{R}{\alpha V}(1 - \alpha V)$

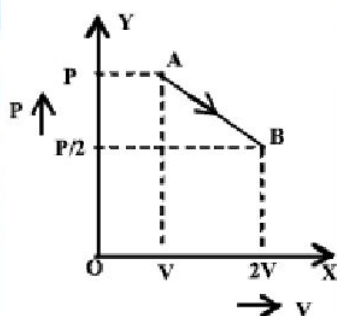
C) $C_v - \frac{R}{2\alpha V}(1 + \alpha V)$

D) $C_v + \frac{R}{2\alpha V}(1 + \alpha V)$

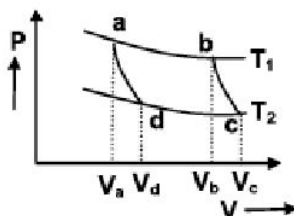
5. A cylinder made of perfectly conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27^\circ$ and pressure $P_0 = 1 \text{ atm}$. Now if the gas in one of the parts is slowly heated to $t = 57^\circ \text{C}$, the distance moved by the piston from the idle of the cylinder will be (length of the cylinder = 84 cm)
- A) 3 cm B) 5 cm C) 2 cm D) 1 cm

MULTIPLE

6. An ideal gas is taken from the state A (pressure P , volume V) to the state B (pressure $P/2$, volume $2V$) along a straight line path in the P - V diagram. Select the correct statement (s) from the following

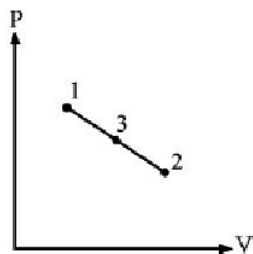


- A) The work done by the gas in the process A to B exceeds the work that would be done by it, if the system were taken from A to B along the isotherm.
- B) In the T - V diagram, the path AB becomes a part of parabola.
- C) In the P - T diagram, the path AB becomes a hyperbola.
- D) In going from A to B, then temperature T of the gas first increase maximum value and then decrease.
7. Two adiabatic processes be and for the same gas are given to intersect two isotherms at T_1 and T_2 (as shown). Then

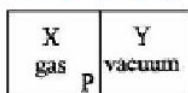


A) $\frac{V_a}{V_b} = \frac{T_2}{T_1}$ B) $\frac{V_a}{V_d} = \frac{T_1}{T_2}$ C) $\frac{V_b}{V_c} = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}}$ D) $V_a V_c = V_b V_d$

8. The diagram shown depicts three different state 1, 2 & 3 for an ideal gas. It is also known that temperatures at states (1) and (2) are same. Then :

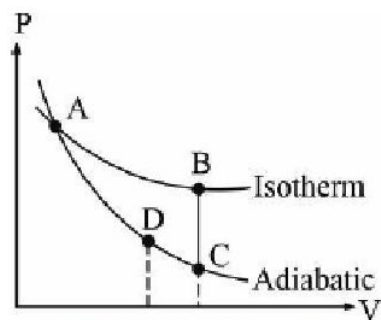


- A) $1 \rightarrow 2$ is an isothermal expansion process
 B) $1 \rightarrow 3$ is an expansion process and is accompanied by cooling
 C) $3 \rightarrow 2$ is an expansion process accompanied by cooling
 D) internal energy must increase in process $1 \rightarrow 3$
9. 2 moles of He are mixed with 2 moles of H_2 in a closed adiabatic container. Initially the mixture occupies 3 litres at $27^\circ C$. The volume is suddenly decreased to $\left(\frac{3}{2}\right)$ litres. Choose the correct option (s) (H_2 and He can be treated as ideal gases) :
- A) γ for mixture is $\frac{3}{2}$
 B) final temperature = $300\sqrt{2}K$
 C) C_p for mixture $2R$
 D) work done in compression is totally converted into internal energy
10. A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y . Which of the following statements is correct ?

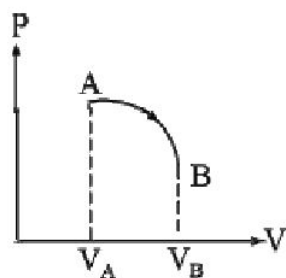


- A) No work is done by X B) X decreases in temperature
 C) X increases in internal energy D) X pressure becomes half

11. Figure shows various process in the PV – plane for an ideal gas. The process ADC is an adiabatic, AB is an isotherm, and BC is a constant – volume process. The heat added to the gas along AB is 400 cal. The change in internal energy from C to A is + 1000 cal and the work done from D to C is 150 cal



- A) work done by the gas from A to B is 400 cal
 B) heat added to the system from B to C is – 1000 cal
 C) the change in internal energy from C to D is 150 cal
 D) work done by the gas from D to A is 850 cal
12. During experiment, an ideal gas is found to obey a condition $\frac{P^2}{\rho} = \text{constant}$ [ρ = density of the gas]. The gas is initially at temperature T , pressure P and density ρ . The gas expands such that density changes to $\frac{\rho}{2}$.
- A) The pressure of the gas changes to $\sqrt{2}P$
 B) The temperature of the gas changes to $\sqrt{2}T$
 C) The graph of the above process on the $P - T$ diagram is parabola
 D) The graph of the above process on the $P - T$ diagram is hyperbola
13. One mole of an ideal monoatomic gas is taken through process AB given by $P = \alpha - \beta V^2$ (where α and β are positive constants) on $P - V$ diagram. Which of the following is the correct statement regarding the given process?

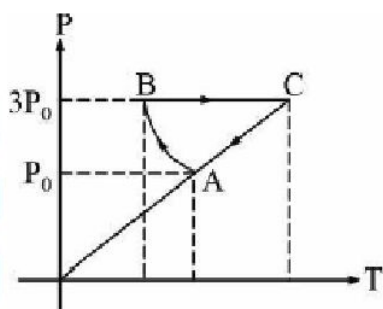


A) Temperature is maximum at $V = \sqrt{\frac{\alpha}{3\beta}}$ B) Temperature is minimum at $V = \sqrt{\frac{\alpha}{3\beta}}$

C) At point A rate of increase of temperature of gas with volume is more than at point B

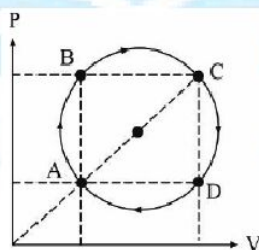
D) Rate of increase of temperature of gas with volume is maximum at $V = \sqrt{\frac{\alpha}{\beta}}$

14. One mole of monoatomic gas is taken through cyclic process shown below. $T_A = 300K$. Process AB is defined as $PT = \text{constant}$.



- A) Work done in process AB is $-400R$
 B) Change in internal energy in process CA is $900R$
 C) Heat transferred in the process BC is $2000R$
 D) Change in internal energy in process CA is $-9000R$

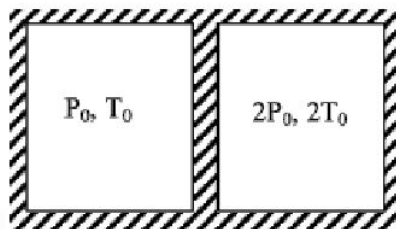
15. A sample of ideal gas of some mass is taken through the cyclic process shown in the figure. The temperature of the gas at state A is $T_A = 200K$, and at state C is $T_C = 1800K$. choose the correct option:



- A) $\frac{V_A}{V_C} = \frac{1}{3}$ B) $\frac{V_A}{V_B} = \frac{1}{9}$ C) $\frac{P_A V_A}{P_C V_C} = \frac{1}{3}$ D) $\frac{P_A V_A}{P_C V_C} = \frac{1}{9}$

PARAGRAPH – 16 – 18

A conducting piston divides adiabatic cylinder of volume V_0 into two equal parts as shown in the figure. Both parts contain ideal monatomic gases. The initial pressure and temperature of gas in left compartment are P_0 and T_0 while that in right compartment are $2P_0$ and $2T_0$. Initially the piston is kept fixed and the system is allowed to acquire a state of thermal equilibrium.



16. The pressure in left compartment after thermal equilibrium is achieved is :
- A) P_0 B) $\frac{3}{2}P_0$ C) $\frac{4}{3}P_0$ D) None of these
17. The heat that flown from right compartment to left compartment before thermal equilibrium is achieved is :
- A) P_0V_0 B) $\frac{3}{4}P_0V_0$ C) $\frac{3}{8}P_0V_0$ D) $\frac{2}{3}P_0V_0$
18. If the pin which was keeping the piston fixed is removed and the piston is allowed to slide slowly such that a state of mechanical equilibrium is achieved. The volume of left compartment when piston is in equilibrium.
- A) $\frac{3}{4}V_0$ B) $\frac{V_0}{4}$ C) $\frac{V_0}{2}$ D) $\frac{2}{3}V_0$

PARAGRAPH – 19 – 20

A piston enclose some diatomic gas in the cylindrical vessel with horizontal longitudinal axis as shown in the drawing. The initial pressure of the air is equal to the external atmospheric pressure of 10^5 Pa . The cross-sectional area of the piston is 0.03 m^2 . An originally un-stretched spring with spring constant 2000 N/m is attached to the piston. The walls of the vessel and the piston are perfectly insulated. The initial volume of the enclosed air is 0.024 m^3 and its initial temperature is 300 K . The air is heated to 360 K with a heating filament built into the vessel.



19. Find the displacement of the piston caused by the heating.
 A) 0.1 m B) 0.2 m C) 0.5 m D) 0.3 m
20. Find the energy delivered by the heating filament.
 A) 1000 J B) 1200 J C) 1400 J D) 1510 J

PARAGRAPH _ 21 – 23

An ideal gas with the ratio of its specific heats = γ , undergoes a process in which its internal energy U is given by $U = aV^\alpha$ where a and α are constants.

21. If the internal energy of the gas increases by ΔU , find the amount of heat ΔQ , needed to be given :

A) $\Delta Q = \Delta U$ B) $\Delta Q = \frac{\gamma \Delta U}{\alpha}$
 C) $\Delta Q = \Delta U \left[1 + \frac{\gamma}{\alpha} \right]$ D) $\Delta Q = \Delta U \left[1 + \frac{\gamma - 1}{\alpha} \right]$

22. The work ΔW performed by the gas when its internal energy increased by ΔU :

A) $\Delta W = \Delta U$ B) $\Delta W = \frac{\gamma \Delta U}{\alpha}$ C) $\Delta W = \frac{(\gamma - 1) \Delta U}{\alpha}$ D) $\Delta W = 0$

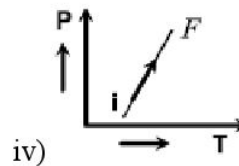
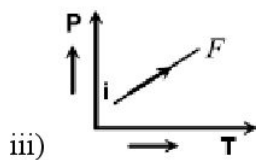
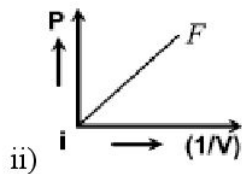
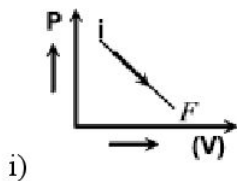
23. The molar specific heat C of the gas during the process :

A) $C = \frac{R}{\gamma - 1}$ B) $C = \frac{R}{\alpha}$ C) $C = \frac{\gamma C_v}{\alpha}$ D) $C = \frac{R}{\gamma - 1} + \frac{R}{\alpha}$

MATRIX MATCHING

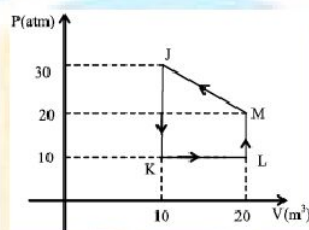
Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) and (D) whereas statements in Column 2 are labelled as p, q, r, s and T. more than one choice from Column 2 can be matched with Column 1.

24. The diagrams below depict different processes for a given amount of an ideal gas. Match Column – I and II.



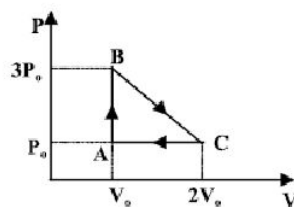
Column – 1		Column – 2	
A)	In fig (i) as the system proceeds from the initial state to the final state	p)	Temperature will remain constant
B)	In Fig (ii), as the system proceeds from the initial state to the final state	q)	Pressure will decrease
C)	In fig (iii), as the system proceeds from the initial state to the final state	r)	Volume must increase
D)	In fig. (iv), as the system proceeds from the initial state to the final state	s)	Temperature may increase, may decrease or may first increase and then decrease

25. Heat given to process is positive, match the following option of column I with the corresponding option of column - II



Column – 1		Column – 2	
A)	JK	P)	$\Delta W > 0$
B)	KL	Q)	$\Delta Q < 0$
C)	LM	R)	$\Delta W < 0$
D)	MJ	S)	$\Delta Q > 0$

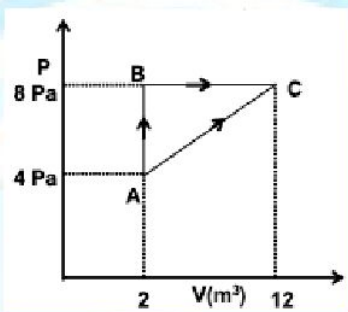
26. One mole of an ideal monatomic gas is taken round the cyclic process ABCA as shown in the figure. Calculate :



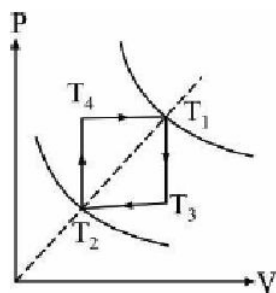
COLUMN – 1		COLUMN – 2	
A)	The work done by the gas	p)	$5/2 P_0 V_0$
B)	The heat rejected by the gas in the path CA;	q)	$\frac{25 P_0 V_0}{8R}$
C)	The net heat absorbed by the gas in the path BC	r)	$P_0 V_0$
D)	The maximum temperature attained by the gas during the cycle.	s)	$\frac{P_0 V_0}{2}$

INTEGER TYPE

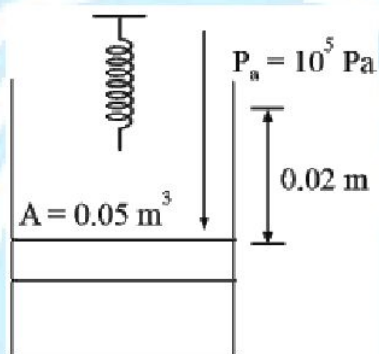
27. In the given figure, an ideal gas changes its state from state A to state C by two paths ABC and AC. The internal energy of gas at A is 30 J and the amount of heat supplied to change its state to C through path AC is 210 J. if the internal energy of gas at C is U then the value of $U/90$ (In J) is_____.



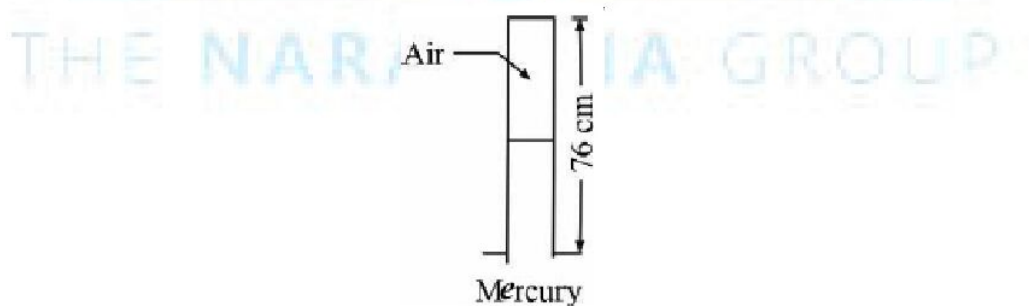
28. One mole helium in a vessel gets the heat from outside and starts expanding to make its volume 2 times the original volume. The heat capacity of the gas in this process is constant and is $\frac{R}{2}$. What is the final temperature of gas (in K)? Initial temperature is 200 K and initial pressure is 40 kPa.
29. A cylindrical vessel is divided in two parts by a fixed partition which is perfectly heat conducting. The wall and piston are thermally insulated from surroundings. The left side contains 0.5 moles of gas with $C_v = 2R$ at temperature of 300 K. The right side contains 4 moles of mixture of gas with $C_v = 1.75R$ at same temperature of 300 K. The piston compresses slowly the right side from volume of V_0 to $\frac{V_0}{4}$. find the total change in internal energy of gases. If $\Delta U = n \times 10^4 J$, fill n in OMR sheet. (Take $R = \frac{25}{3}$ S.I. unit)
30. Between two isotherm we have cycle as shown. Find the work done by the gas during the cycle. (in J) [Take $T_1 = 127^\circ C$, $T_2 = 16^\circ C$, $n = 1$ mole]



31. A cylinder is fitted with a piston as shown in figure contains 0.001 m^3 gas at a pressure of 10^5 Pa . The cross-sectional area of the piston is 0.05 m^2 . Initially the spring does not touch the piston but atmospheric pressure of 10^5 Pa acts on the piston. The gas is slowly heated till the volume is increased to three times the original value. If the force constant of the spring is 25 kN/m , calculate the work done (in joule) by the gas.



32. A glass tube has fixed length 76 cm above the mercury level. There is some space above mercury column in the tube that contains 10^{-3} moles of air at 300 K . Atmospheric pressure remains constant at 76 cm of Hg. If the temperature of the air column slowly decreases by 10 K , find the net heat lost by the air column. Given $\gamma_{\text{air}} = 1.4$. Neglect the effect of surface tension. [Take $R = 25/3 \text{ J/mol-K}$]



33. A monoatomic gas undergoes a process in which it uses 50% of heat in work done by gas. If the process equation of gas is $VP^k = \text{constant}$ then value of k will be.

GASEOUS STATE & THERMODYNAMICS

KEY

1	2	3	4	5	6	7	8	9	10
A	C	B	B	C	ABD	CD	CD	ABD	AD
11	12	13	14	15	16	17	18	19	20
ABC	BD	AC	ACD	AD	B	C	C	A	D
21	22	23	24	25	26	27	28	29	30
D	C	D	A-q,r,s B-p C-r D-s	A-q B-p,s C-s D-q,r	A-r B-p C-s D-q	2	100	2	75
31	32	33							
205	0.25	3							

HINTS

1. ab and cd are adiabatic

$$bc \text{ and } cd \text{ are isothermal. Hence } \left. \begin{array}{l} P_b V_b = P_c V_c \\ P_d V_d = P_a V_a \\ P_c V_c^\gamma = P_d V_d^\gamma \\ P_a V_a^\gamma = P_b V_b^\gamma \end{array} \right\} \Rightarrow \frac{P_b}{P_c} = \frac{P_a}{P_d}$$

2. $\delta = \frac{\Delta V}{V \Delta T}$ from $PV = nRT \Rightarrow P \Delta V + V \Delta P = nR \Delta T$

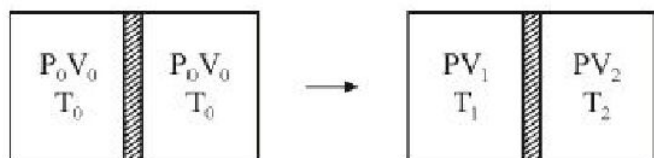
Since ' P ' is constant $\Rightarrow P \Delta V = nR \Delta T \Rightarrow \delta = \left(\frac{nR}{P} \right) \cdot \frac{1}{T}$

3. At constant pressure $Q = nC_p \Delta T$; also $Q = \Delta U + w \Rightarrow nC_p \Delta T = nC_v \Delta T + w$
 $\Rightarrow n(C_p - C_v) \Delta T = w \Rightarrow nR \Delta T = 25$

$$\Delta U = n \frac{6}{2} R \Delta T = 75J \text{ \& } Q = 25 + 75 = 100J$$

4. $Q = \Delta U + W \Rightarrow C = C_v + \frac{RT}{V} \frac{dV}{dT} \Rightarrow C = C_v - \frac{R}{\alpha V} (1 - \alpha V)$

5.



$$2 \frac{P_0 V_0}{T_0} = \frac{PV_1}{T_1} + \frac{PV_2}{T_2} \text{ and } V_1 + V_2 = 2V_0$$

$$V_1 = x_1 A, V_2 = x_2 A \Rightarrow x_1 = 44 \text{ cm and } x_2 = 40 \text{ cm}$$

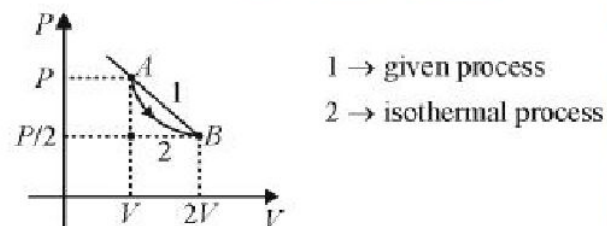
6. from the graph we can collude that

$$W_{\text{given process}} > W_{\text{isothermal process}}$$

For AB process

$$P = -kV + C \Rightarrow T = -\left(\frac{k}{nR}\right)V^2 + CV$$

Which is the equation of parabola



7. For A, D $T_1 V_a^{\gamma-1} = T_2 V_d^{\gamma-1} \Rightarrow \left(\frac{V_a}{V_d}\right)^{\gamma-1} = \frac{T_2}{T_1}$

For B, C $T_1 V_b^{\gamma-1} = T_2 V_c^{\gamma-1} \Rightarrow \left(\frac{V_b}{V_c}\right)^{\gamma-1} = \frac{T_2}{T_1} \Rightarrow \frac{V_b}{V_c} = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}}$

So $\left(\frac{V_a}{V_d}\right) = \left(\frac{V_b}{V_c}\right)$, is., choices C and D are correct. Hence choice A and B are wrong.

8. $P = -mV + c \Rightarrow T = -\left(\frac{m}{nR}\right)V^2 + \left(\frac{c}{nR}\right)V \Rightarrow T_3 > T_1 = T_2$

∴ 1 → 3 → expansion process accompanied by heating

3 → 2 → expansion process accompanied by cooling

9. Degree of freedom of $He = 3$

Degree of freedom of $H_2 = 5$

$$\text{Average degree of freedom} = \frac{3 \times 2 + 5 \times 2}{2 + 2} = 4$$

Now $\gamma = 1 + \frac{2}{f} = \frac{3}{2}$; For adiabatic process $\Delta Q = 0$ and $PV^\gamma = \text{const}$ $TV^{\gamma-1} = \text{const}$

From first law $\Delta U = \Delta W$ [$\because \Delta Q = 0$]

10. This process is free expansion so

$$\Delta Q = \Delta U = \Delta W = 0 \quad \text{and} \quad P_1 V_1 = P_2 V_2$$

Only initial and final points are defined, process in between is not defined.

11. $\Delta Q = \Delta U + \Delta W$

For process A to B, $\Delta Q = \Delta U$

For process B to C, $\Delta Q = \Delta U$

For process C to D, $\Delta U = -\Delta W$

For process D, to A, $\Delta U = -\Delta W$

12. $\frac{P^2}{\rho} \Rightarrow \frac{P^2 RT}{PM} = k \Rightarrow PT = \left(\frac{kM}{R} \right) \dots (i)$

$$\frac{P^2}{\rho} = \frac{P^2}{\rho} \Rightarrow P' = \frac{P}{\sqrt{2}}$$

Hence from eq. (i) $T' = T\sqrt{2}$

$PT = \text{constant}$ hence P - T curve is a hyperbola.

13. $PV = RT$

$$(\alpha - \beta V^2)V = RT; \quad T = \frac{\alpha V}{R} - \frac{\beta V^2}{R}; \quad \frac{dT}{dV} = \frac{\alpha}{R} - \frac{2\beta V}{R}$$

For maximum value of T; $\frac{dT}{dV} = 0 \Rightarrow V = \sqrt{\frac{\alpha}{2\beta}}$

14. (a) Process AB: $PT = \text{Constant}$

$$\Rightarrow \frac{nRT^2}{V} = \text{constant}; \quad W = \int_A^B P dV = \int_A^B \frac{\text{Constant}}{T} dV$$

$$\Rightarrow \frac{dV}{dT} = \frac{2nRT}{\text{Constant}}; \quad W = \int_{300}^{100} \frac{\text{Constant}}{T} \cdot \frac{2nRT}{\text{Constant}} dt$$

$$\Rightarrow \frac{P_A}{P_B} = \frac{T_B}{T_A} \Rightarrow \frac{1}{3} = \frac{T_B}{T_A} \Rightarrow T_B = \frac{300}{3} = 100$$

$$\Rightarrow W = 2nR(100 - 300) \Rightarrow W_{AB} = -400nR$$

(b) Process CA : Isochoric $\frac{P}{T}$ constant :

$$\frac{T_A}{T_C} = \frac{P_A}{P_C}; \quad \frac{T_A}{T_C} = \frac{P_A}{P_B}$$

$$\Rightarrow \frac{T_A}{T_C} = \frac{1}{3} \Rightarrow T_C = 3T_A$$

$$T_C = 900R \Rightarrow \Delta U = nC_V \Delta T = (1) \frac{3}{2} R \times (T_A - T_C) = \frac{3}{2} R \times (300 - 900)$$

$$\Rightarrow |\Delta U| = (900R)$$

(c) Process BC: Isobaric $Q = nC_P \Delta T$

$$\Rightarrow q = (1) \frac{5}{2} R \times (T_C - T_B); Q = \frac{5}{2} R \times (900 - 100) \Rightarrow Q = \frac{5}{2} R \times 800 \Rightarrow Q = 2000R$$

15. Point A and C are on the same line passing through origin

$$\Rightarrow \frac{P_A}{V_A} = \frac{P_C}{V_C} \dots\dots(i)$$

Also $T_A = 200K = \frac{P_A V_A}{nR}$ and also $T_C = 1800K = \frac{P_C V_C}{nR}$

$$\Rightarrow \frac{P_A V_A}{P_C V_C} = \frac{1}{9} \dots(ii)$$

From eq. (i) and (ii) $\frac{V_A}{V_C} = \frac{1}{3}$

16. Heat gain by left part = heat lost by right part

$$\Rightarrow \frac{3}{2} nR (T - T_0) = \frac{3}{2} nR (2T_0 - T) \Rightarrow T = \frac{3T_0}{2}$$

Let final pressure = $p \Rightarrow \frac{p}{T} = \frac{P_0}{T_0} \Rightarrow p = \frac{3P_0}{2}$

17. Heat flow = $\frac{3}{2} nR \left(\frac{3T_0}{2} - T_0 \right) = \frac{3}{4} nRT_0 = \frac{3}{4} P_0 \frac{V_0}{2} = \frac{3}{8} P_0 V_0$

18. P (final in left) = $\frac{3P_0}{2} = P$ (final in Right part). So when pin is removed, piston will not move.

19-20. $\frac{p_1 V_1}{T_1} = \frac{\left(p_1 + \frac{kx}{A} \right) (V_1 + Ax)}{T_2}; kx^2 + \left(p_1 A + \frac{kV_1}{A} \right) x + \left(p_1 V_1 - \frac{p_1 V_2 T_2}{T_1} \right) = 0$

$$2000x^2 + 4600x - 480 = 0; x = 0.1m$$

$$W_{gas} + W_{atmosphere} + W_{spring} = 0; W_{gas} - P_e Ax - \frac{1}{2} kx^2 = 0$$

$$W_{gas} = P_e Ax - \frac{1}{2} kx^2 = 310J; Q = \Delta U + p_e Ax + \frac{1}{2} kx^2; \Delta U = \frac{f}{2} Nk \Delta T$$

$$\Delta U = \frac{f}{2} \frac{p_1 V_1}{T_1} \Delta T = \frac{5}{2} \frac{10^5 Pa \times 0.024m^3}{300K} \cdot 60K - 1200.0J$$

21. Conceptual

22. Conceptual

23. Since $U = aV^\alpha$

$$\Delta U = C_V \Delta T = (a\alpha) V^{(\alpha-1)} \Delta V \quad (i)$$

Where ΔV is the change in volume during the process or $\Delta U = \alpha \left[\frac{aV^\alpha}{V} \right] \Delta V = \frac{\alpha U}{V} \Delta V$

This gives $\frac{\Delta U}{U} = \frac{\alpha \Delta V}{V} \dots(ii)$

Work done during the process is $\Delta W = P\Delta V$; i.e., $\Delta W = P \left[\frac{V\Delta U}{U\alpha} \right] = \frac{PV}{\alpha} \left[\frac{\Delta U}{U} \right] = \frac{\Delta U}{\alpha} \left[\frac{PV}{U} \right]$

But $PV = nRT = nC_v(\gamma-1)T$; $= (\gamma-1)[nC_vT] = (\gamma-1)U$

Hence $\Delta W = \left(\frac{\gamma-1}{\alpha} \Delta U \right)$

Therefore $\Delta Q = \Delta U + \Delta W = \Delta U \left[1 + \frac{\gamma-1}{\alpha} \right] \dots (iii)$

Let C be the molar specific heat in the process

$$\Delta Q = C\Delta T = \Delta U \left[1 + \frac{\gamma-1}{\alpha} \right] = C_v\Delta T \left[1 + \frac{\gamma-1}{\alpha} \right] \Rightarrow C = C_v + \frac{C_v(\gamma-1)}{\alpha} = \left(\frac{R}{\gamma-1} \right) + \frac{R}{\alpha}$$

24. Conceptual

25. Temperature at :

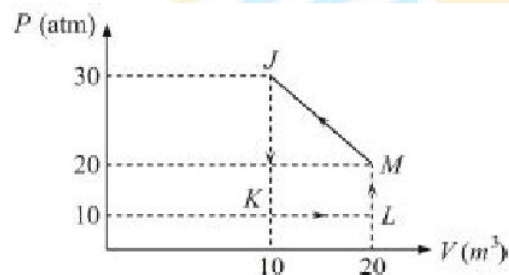
$$J = \frac{30 \times 10}{nR}, K = \frac{10 \times 10}{nR}, L = \frac{20 \times 10}{nR}, M = \frac{20 \times 20}{nR}$$

For JK : $W = 0$; $\Delta U < 0 \Rightarrow \Delta Q < 0$

For KL : $W = 10 \times 10$; $\Delta U > 0 \Rightarrow \Delta Q > 0$

For LM : $W = 0$; $\Delta U > 0 \Rightarrow \Delta Q > 0$

For MJ : $W < 0$; $\Delta U < 0 \Rightarrow \Delta Q < 0$



26. $W = \text{Area of triangle} = \frac{1}{2} \times 2P_0 \times V_0 = P_0V_0$

For CA : $W = -P_0V_0 \Rightarrow \Delta U = \frac{3}{2}R \left[\frac{P_0V}{R} - \frac{2P_0V_0}{R} \right] = -\frac{3}{2}P_0V_0$ & $\Delta Q = -P_0V_0 - \frac{3}{2}P_0V_0 = -\frac{5}{2}P_0V_0$

For BC : $W = \frac{1}{2} [3P_0 + P_0]V_0 = 2P_0V_0 \Rightarrow \Delta U = \frac{3}{2}R \left[\frac{2P_0V}{R} - \frac{3P_0V_0}{R} \right] = -\frac{3}{2}P_0V_0$ & $\Delta Q = \frac{P_0V_0}{2}$

27. Process A to C

$Q = 210 \text{ J}$

Work done $W_{AC} = \text{area under } AC = (10 \times 4) + \left(\frac{10 \times 4}{2} \right) = 60 \text{ J}$

From 1st law of thermodynamics.

$$\Delta U = Q - W_{AC} \Rightarrow U_C - U_A = 210 - 60 \quad \therefore U_C = U_A + 150 = 30 + 150 = 180 \text{ J}$$

28. Process is polytropic $C = C_v - \frac{R}{m-1}$

$$\frac{R}{2} = \frac{3}{2}R - \frac{R}{m-1} \Rightarrow m = 2$$

$$PV^2 = C$$

$$40 \times V_0^2 = P (2V_0)^2$$

$$P = 10 \text{ kPa}$$

$$\frac{PV}{T} = \frac{P_0 V_0}{T_0} \Rightarrow \frac{10 \times 2V_0}{T} = \frac{40 \times V_0}{200} \Rightarrow T = 100\text{K}$$



29. $n_1 C_{v1} dT + n_2 C_{v2} dT + PdV = 0$

$$\frac{1}{2} \times 2RdT + 4 \times \frac{7R}{4} dT + 4RT \frac{dV}{V} = 0 ; 2 \int \frac{dT}{T} + \int \frac{dV}{V} = 0$$

$$2 \ln \left(\frac{T}{300} \right) + \ln \left(\frac{1}{4} \right) = 0 ; \ln \frac{T}{300} = \ln 2$$

$$\Delta U = \frac{1}{2} \times 2R \times (600 - 300) + 4 \times \frac{7R}{4} (600 - 300) = 8R \times 300 = 8 \times \frac{25}{3} \times 300 = 2 \times 10^4 J$$

30. $\frac{P_2}{P_1} = \frac{V_2}{V_1} \Rightarrow P_2 V_1 = P_1 V_2$

$$W = P_2 (V_2 - V_1) - P_1 (V_2 - V_1)$$

$$W = nRT_2 + nRT_1 - P_2 V_1 - P_1 V_2 \quad \text{and} \quad P_2 = \frac{nRT_2}{V_2}, P_1 = \frac{nRT_1}{V_1}$$

$$\Rightarrow P_2 V_1 = P_1 V_2 = nR \sqrt{T_1 T_2} \Rightarrow W = nR (\sqrt{T_1} - \sqrt{T_2})^2 = 1 \times \frac{25}{3} (20 - 17)^2 = 75 J$$

31. $W_{gas} + W_{spring} + W_{atm} = 0$

$$W_{gas} - \frac{1}{2} \times 25 \times 10^3 \times (0.02)^2 - 10^5 \times 0.05 \times 0.04 = 0 ; W_{gas} = 205 J$$

32. Pressure of air (P) = $\rho_{Hg} g (76 - x)$

$$\frac{P}{V} = \frac{\rho_{Hg} g}{A} = \text{constant}$$

Molar heat capacity (C)

$$= \frac{R}{\gamma - 1} - \frac{R}{n - 1}$$

$$= \frac{R}{1.4 - 1} - \frac{R}{-1 - 1} = 3R$$

$$\therefore \Delta Q = nC\Delta T = (10^{-3}) \left(3 \times \frac{25}{3} \right) (10) = 0.25 J$$

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33. $VP^k = \text{constant} \Rightarrow PV^{\frac{1}{k}} = \text{constant} : n = \frac{1}{k}$

50% heat as work

$$dQ = dW + dU ; dU = dW + \frac{dQ}{2} = \frac{f}{2} nR\Delta T ; dQ = 2dU = fnR\Delta T = nC\Delta T$$

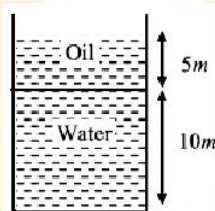
$$C = fR = 3R$$

Now :

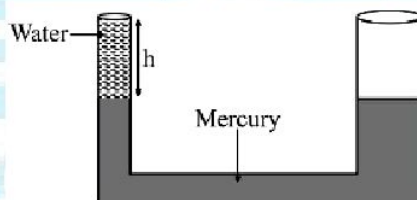
$$C = \frac{R}{\gamma - 1} - \frac{R}{n - 1} ; 3R \frac{R}{\frac{5}{3} - 1} - \frac{R}{n - 1} \Rightarrow n = \frac{1}{3} \Rightarrow k = \frac{1}{n} = 3$$

LIQUIDS SINGLE

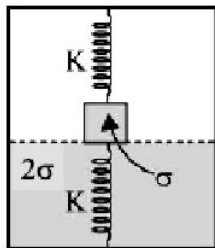
1. A vessel of uniform cross section (open at the top with an orifice at the bottom) contains oil (relative density 0.8) on top of water. Water flows out of the vessel. The initial speed of water for the values given in the figure is nearly. (Take $g=10 \text{ m/s}^2$)



- A) 10.0 m/s B) 8.0 m/s C) 16.7 m/s D) 5.0 m/s
2. Two communicating vessels contain mercury. The diameter of one vessel is n times than the diameter of the other. A column of water of height h is poured into the left vessel. The mercury level will rise in the right-hand vessel (S = relative density of mercury and ρ = density of water) by

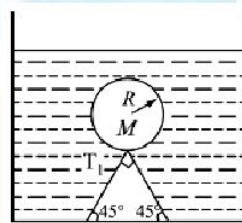


- A) $\frac{n^2 h}{(n+1)^2 s}$ B) $\frac{h}{(n^2 + 1)s}$ C) $\frac{h}{(n+1)^2 s}$ D) $\frac{h}{n^2 s}$
3. A cubic block of side a is connected with two similar vertical springs as shown. Initially, bottom surface of the block of density σ touches the surface of the fluid of density 2σ while floating. A weight is placed on the block so that it is immersed half the fluid, then the weight is :

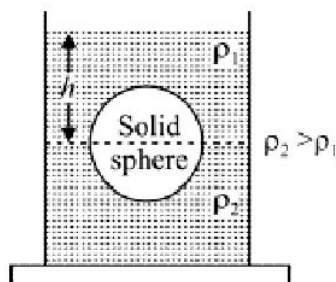


- A) $a \left(\frac{K}{2} + a^2 \sigma g \right)$ B) $a(K + a^2 \sigma g)$
 C) $a \left(K + \frac{a^2}{2} \sigma g \right)$ D) $\frac{a}{2}(K + a^2 \sigma g)$

4. A hollow sphere of mass M and radius R is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of the tank by two wires which makes angle 45° with the horizontal as shown in the figure. The tension T_1 in the wire is :

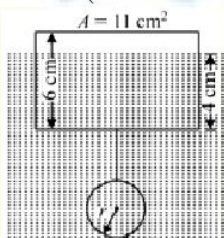


- A) $\frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{\sqrt{2}}$ B) $\frac{2}{3}\pi R^3 \rho_w g - Mg$
 C) $\frac{4}{3}\pi R^3 \rho_w g - Mg$ D) $\frac{4}{3}\pi R^3 \rho_w g + Mg$
5. A body of density ρ' is dropped from rest at a height h into a lake of density ρ where $\rho > \rho'$. Neglecting all dissipative forces calculate the maximum depth to which the body sinks before returning to float on the surface.
- A) $\frac{h}{\rho - \rho'}$ B) $\frac{h\rho'}{\rho}$ C) $\frac{h\rho'}{\rho - \rho'}$ D) $\frac{h\rho}{\rho - \rho'}$
6. A solid sphere of radius r is floating of two immiscible liquids of densities ρ_1 and ρ_2 ($\rho_2 > \rho_1$), half of its volume lying in each. The height of the upper liquid column from the interface of the two liquids is h . The force exerted on the sphere by the upper liquid is (atmospheric pressure $= p_0$ and acceleration due to gravity is g) :

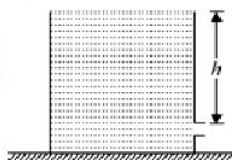


- A) $P_0 \pi r^2 + \left(h - \frac{2}{3}r\right) \pi r^2 \rho_1 g$ B) $\left(h - \frac{2}{3}r\right) \pi r^2 \rho_1 g$
 C) $\frac{2}{3} \pi r^3 \rho_1 g$ D) $P_0 \times \pi r^2$

7. Figure shows a metal ball suspended by thread of negligible mass from an upright cylinder that floats partially submerged in water. The cylinder has height 6 cm, face area 11 cm^2 on the top and bottom and density 0.5 g/cm^3 . 4 cm of cylinder's height is inside the water surface. If density of the metal ball is 8 gm/cm^3 then its radius is equal to: ($\rho_w = 1 \text{ gm/cm}^3$)



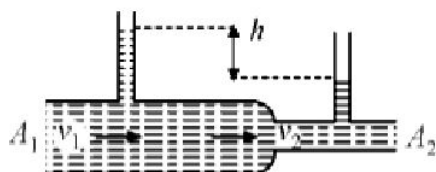
- A) $\left(\frac{3}{8}\right)^{1/3} \text{ cm}$ B) $\left(\frac{3}{4}\right)^{1/3} \text{ cm}$ C) $\left(\frac{6}{8}\right)^{1/3} \text{ cm}$ D) $\left(\frac{5}{11}\right)^{1/3} \text{ cm}$
8. In the figure shown, a light container is kept on a horizontal rough surface of coefficient of friction $\mu = \frac{Sh}{V}$. A very small hole of area S is made at depth 'h'. Water of volume 'V' is filled in the container. The friction is not sufficient to keep the container at rest. The acceleration of the container initially is:



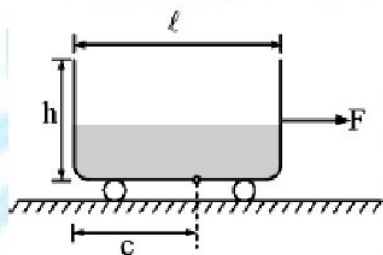
- A) $\frac{V}{Sh} g$ B) g C) zero D) $\frac{Sh}{V} g$

MULTIPLE

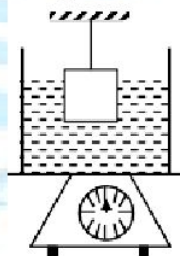
9. A liquid flows through a horizontal tube. The velocities of the liquid in the two sections, which have areas of cross section A_1 and A_2 are v_1 and v_2 respectively. The difference in the levels of the liquid in the two vertical tubes is h. Then



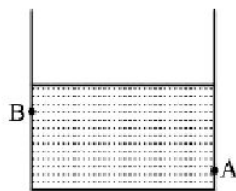
- A) The volume of the liquid flowing through the tube in unit time is $A_1 v_1$
- B) $v_2 - v_1 = \sqrt{2gh}$
- C) $v_2^2 - v_1^2 = 2gh$
- D) The energy per unit mass of the liquid is the same in both sections of the tube
10. A rectangular vessel of dimension $(l \times b \times h)$ and mass M contains a liquid of density ρ . The vessel has an orifice at its bottom at a distance C from the rear wall as shown in the figure.



- A) The maximum volume of the water that can be stored when the vessel is accelerated is $hcb / 2$
- B) The maximum volume of the water that can be stored when the vessel is accelerated is $hbl / 2$
- C) Force F that must be applied when maximum water stored is $\left[M + \frac{hcb\rho}{2} \right] \frac{hg}{c}$
- D) Force F that must be applied when maximum water stored is $\left[M + \frac{hcb\rho}{2} \right] \frac{lg}{c}$
11. The mass of block is m_1 and that of liquid with the vessel is m_2 . The block is suspended by a string (tension T) partially in the liquid. The reading of the weighing machine placed below the vessel.



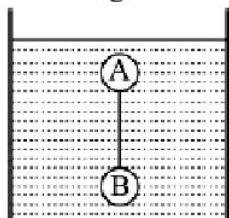
- A) is $(m_1 + m_2)g$
- B) is greater than $(m_1 + m_2)g$
- C) is equal to $(m_1 g + m_2 g - T)$
- D) is less than $(m_1 + m_2)g$
12. The container carrying some liquid shown in the diagram is given some small acceleration $\vec{a}$.



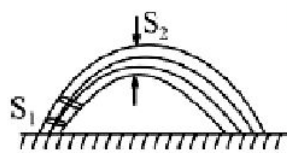
- A) If $\vec{a}$ is directed upwards, $P_A - P_B$ increases

- B) If $\vec{a}$ is directed towards right, $P_A - P_B$ decreases
 C) If $\vec{a}$ is directed downwards, $P_A - P_B$ remain same
 D) If $\vec{a}$ is directed towards left, $P_A - P_B$ remain same

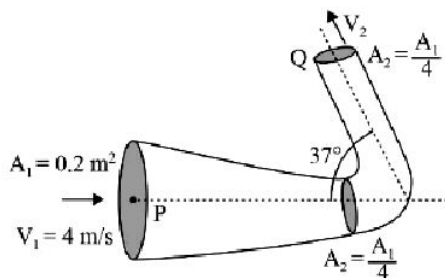
13. Two solid spheres A and B of equal volumes but of different densities d_A and d_B are connected by a string. They are fully immersed in a fluid of density d_F . They get arranged into an equilibrium state as shown in the figure with a tension in the string. The arrangement is possible only if :



- A) $d_A < d_F$ B) $d_B > d_F$ C) $d_A > d_F$ D) $d_A + d_B = 2d_F$
14. Water jet is projected at an angle to the horizontal. At the point of projection, the area of the jet is S_1 and at the highest point, the area of the jet is S_2 . The initial velocity of projection is u .

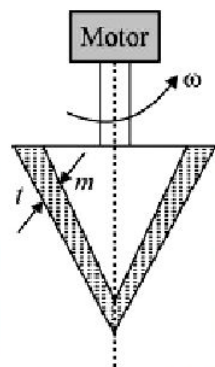


- A) The angle of projection is $\cos^{-1} \left(\frac{S_1}{S_2} \right)$
 B) The range on the level ground is $\frac{2u^2}{g} \frac{S_1}{S_2} \sqrt{1 - \frac{S_1^2}{S_2^2}}$
 C) The maximum height reached from the ground is $\frac{2u^2}{g} \left(1 - \frac{S_1^2}{S_2^2} \right)$
 D) The rate of volume flow is $S_2 u$
15. Oil enters the bend of a pipe in the horizontal plane with velocity 4 ms^{-1} and pressure $280 \times 10^3 \text{ Nm}^{-2}$ as shown in figure. (Take specific gravity of oil as 0.9 and $\sin 37^\circ = 0.6$)



- A) the pressure of oil at the point Q , $(P_2) = 172 \times 10^3 \text{ N/m}^2$

- B) the pressure of oil at the point $Q, (P_2) = 388 \times 10^3 \text{ N/m}^2$
 C) the force required to hold the bend in the place is $63 \times 10^3 \text{ N}$
 D) the force required to hold the bend in the place is $76 \times 10^3 \text{ N}$
16. In the figure shown, there is a conical shaft rotating on a bearing of very small clearance t . The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η . The shaft is having radius R and height h . If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω . Then :



A) $P = \frac{\pi \omega^2 \eta R^3 \sqrt{R^2 + h^2}}{2t}$

B) $\tau = \frac{\pi \omega \eta R^3 \sqrt{R^2 + h^2}}{2t}$

C) $P = \frac{\pi \omega \eta R^3 h}{2t}$

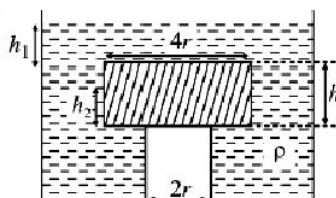
D) $\tau = \frac{\pi \omega \eta R^3 h^2}{2t \sqrt{R^2 + h^2}}$

PASSAGE 17 – 19

A cylindrical tank has a hole of diameter $2r$ in its bottom. The hole is covered with a wooden cylinder block of diameter $4r$, height h and density $\rho/3$.

Process 1 : Initially, the tank is filled with water of density ρ to a height such that the height of water above the top of the block is h_1 (measured from the top of the block).

Process 2 : The water is removed from the tank to a height h_2 (measured from the bottom of the block), as shown in the figure. The height h_2 is smaller than h (height of the block) and thus the block is exposed to the atmosphere.

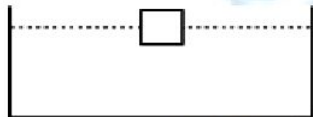


17. Then the minimum value of height h_1 (in process – 1), for which the block just starts to move up is :
- A) $\frac{2h}{3}$ B) $\frac{5h}{4}$ C) $\frac{5h}{3}$ D) $\frac{5h}{2}$

18. The height of the water level h_2 (in process – 2), for which the block remains in its original position without the application of any external force is :
- A) $\frac{h}{3}$ B) $\frac{4h}{9}$ C) $\frac{2h}{3}$ D) h
19. In process 2, if h_2 is further decreased, then :
- A) Cylinder will not move up and remains at its original position
- B) For $h_2 = \frac{h}{3}$, cylinder again starts moving up
- C) For $h_2 = \frac{h}{4}$, cylinder again starts moving up
- D) For $h_2 = \frac{h}{5}$, cylinder again starts moving up

PASSAGE FOR 20 – 22

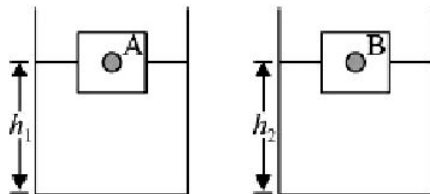
A cube made of wood having specific gravity 0.4 and side length ' a ' is floated in a large tank full of water.



20. Which action would change the depth to which block is submerged ?
- A) more water is added in the tank B) atmospheric pressure increases
- C) the tank is accelerated upwards D) none of these
21. If the cube is depressed slightly, it executes SHM from it's position. What is it's time period?
- A) $2\pi\sqrt{\frac{a}{g}}$ B) $2\pi\sqrt{\frac{5a}{2g}}$ C) $2\pi\sqrt{\frac{2a}{5g}}$ D) $\frac{4\pi}{5}\sqrt{\frac{a}{g}}$
22. What can be the maximum amplitude of it's vertical simple harmonic motion ?
- A) $\frac{a}{2}$ B) $0.4a$ C) $0.6a$ D) $0.2a$

PASSAGE 23 - 25

Two ice cubes of side 10 cm, having cavity of volume 20cm^3 at centre of cube but filled with different materials A and B respectively. The specific gravity of material A is 4.9 and specific gravity of material B is 1.9. Now these cubes are placed in two different vessels of same base area as show in figure. The water level before putting blocks in vessels are same. Assume that ice melts uniformly from all sides and with same constant rate in both the vessels. (specific gravity of ice = 0.9)



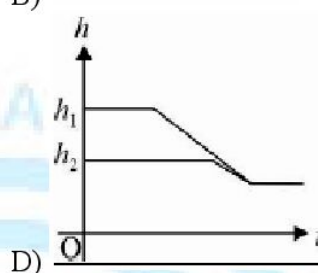
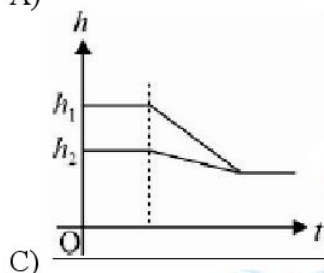
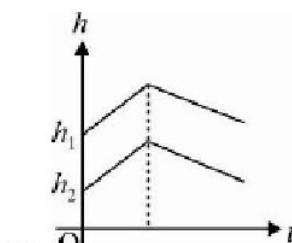
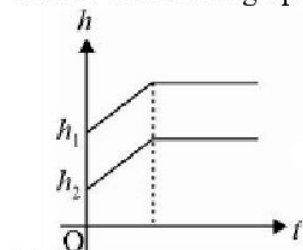
23. Find the ratio of initial submerged volumes of the block containing A and B respectively :
- A) $\frac{49}{46}$ B) $1 : 1$ C) $\frac{49}{19}$ D) $4 : 1$
24. Choose the correct statement :
- A) Both cubes sink simultaneously

B) A and B sink only after complete melting of ice

C) A sinks prior to B

D) B sinks prior to A

25. Choose the correct graph showing the variation of heights of water-level in two vessels with time:



MATRIX MATCHING

26. MATCH THE FOLLOWING :

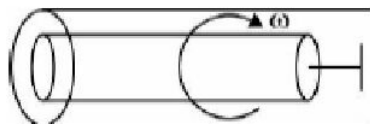
COLUMN – I		COLUMN – II	
P)	Hydrostatic force on the side wall of the cubical vessel, ρ = density of liquid 	1)	$A\rho v^2$
Q)	Buoyant force on the cube, ρ = density of liquid 	2)	$\frac{1}{2}A\rho v^2$
R)	Impact (reaction) force on the vessel by the liquid coming out of the vessel, ρ = density of liquid 	3)	ρghA
S)	$\rho =$ 	4)	$\frac{1}{2}\rho ghA$

Codes :

	P	Q	R	S		P	Q	R	S
A)	4	3	2	1	B)	4	1	1	1
C)	4	3	1	2	D)	4	2	1	3

27. MATCH THE FOLLOWING :

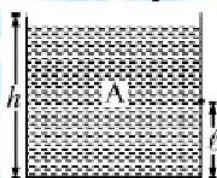
A cylindrical piston of length l and radius R is rotated with constant velocity ω in a cylindrical tube of radius $R+h$ (where $h \ll R$) filled with oil.



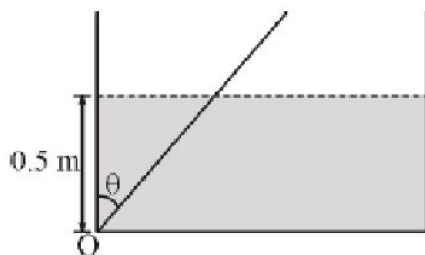
COLUMN – I		COLUMN – II	
A)	Shearing stress	p)	$\frac{2\pi\eta R^2\omega}{h}$
B)	Force	q)	$\frac{2\pi\eta\omega lR^3}{h}$
C)	Torque	r)	$\frac{\eta R\omega}{h}$
D)	Power	s)	$\frac{2\pi\eta\omega^2 lR^3}{h}$

NUMERICAL

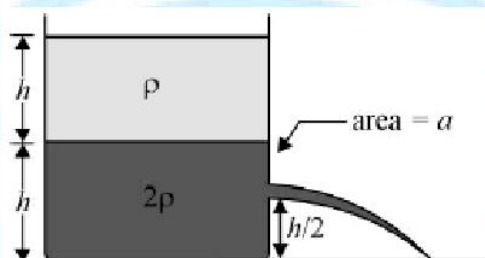
28. The side wall of a wide vertical vessel of height $h = 75\text{cm}$ has a narrow vertical slit running all the way down to the bottom of the vessel. The length of the slit is $l = 50\text{cm}$ and the width $b = 1.0\text{mm}$. With the slit closed, the vessel is filled with water. Find the resultant force of reaction of the water flowing out the vessel immediately after the slit is opened.



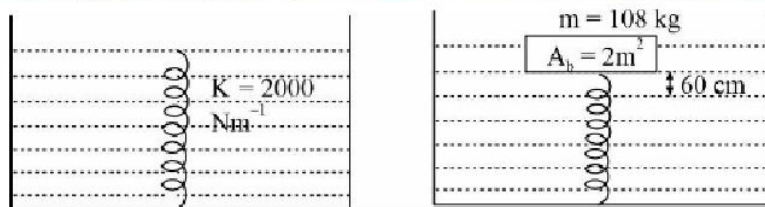
29. A wooden plank of length 1m and uniform cross section is hinged at one end to the bottom of a tank as shown in the figure. The tank is filled with water up to a height of 0.5m . The specific gravity of the plank is 0.5 . If θ is the angle that the plank makes with the vertical equilibrium position. (Exclude the case $\theta = 0$). Find the value of $\frac{1}{\cos^2 \theta}$.



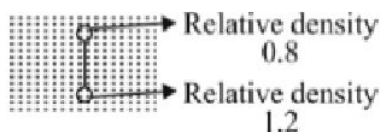
30. A body floats with $(1/3)$ of its volume outside water when submerged in water and $(5/9)$ of its volume outside another liquid when submerged in another liquid. If density of liquid is $n \times 10^3 (kg/m^3)$, find n .
31. A fixed cylindrical tank having large cross-section area is filled with two liquids of densities ρ and 2ρ and in equal heights as shown in the figure. A small hole of area of cross-section $a = \sqrt{6} cm^2$ is made at height $\frac{h}{2}$ from the bottom. Find the area of cross-section of stream of liquid in cm^2 just before it hits the ground.



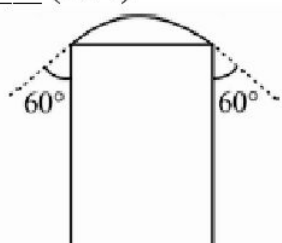
32. A cube (density $0.5 gm/cc$) of side $10 cm$ is floating in water kept in a cylindrical beaker of base area $1500 cm^2$. When a mass m is kept on wooden block the level of water rises in the beaker by $2 mm$. find the mass m (in gm).
33. In a tank of horizontal cross-sectional area $1 m^2$, a spring with force constant $2000 Nm^{-1}$ is fixed in vertical position upto the height of the water as shown in figure 1. A block of mass $180 kg$ is gently placed over the spring and it attains the equilibrium position as shown in figure 2. If base area of the block is $0.2 m^2$ and height $60 cm$, then find compression in the spring in m in equilibrium positions (Take $g = 10 m/s^2$, $\rho_w = 1000 kg/m^3$)



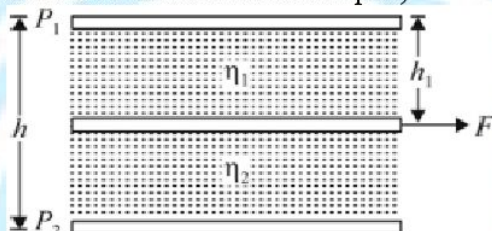
34. Two spheres of volume $250 cc$ each but of relative densities 0.8 and 1.2 are connected by a string and the combination is immersed in a liquid in vertical position as shown in figure. The tension in the string is (in N) ($g = 10 m/s^2$)



35. A soap bubble is being blown on a tube of radius 1 cm. the surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of 60° with the tube as shown. The excess of pressure over the atmospheric pressure in the tube is ____ (in Pa)



36. A thin horizontal movable plate is separated from two fixed horizontal plates P_1 and P_2 by two highly viscous liquids of coefficient of viscosity η_1 and η_2 as shown, where $\eta_2 = 4\eta_1$. Area of contact of movable plate with each fluid is same. If the distance between two fixed plates is h , and the distance of movable plate from upper fixed plate such that the movable plate can be moved with a constant velocity by applying a minimum constant horizontal force F on movable plate is h_1 , then h/h_1 is _____. (assume velocity gradient to be uniform in each liquid)



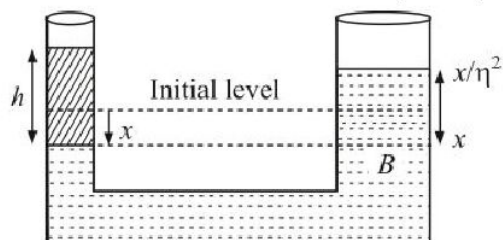
LIQUIDS KEY

1	2	3	4	5	6	7	8	9	10
C	B	B	A	C	A	A	D	ACD	AC
11	12	13	14	15	16	17	18	19	20
CD	AB	ABD	AB	AD	AB	C	B	A	D
21	22	23	24	25	26	27	28	29	30
C	B	A	C	D	C	A-r, B-p, C-q, D-s	5	2	1.50
31	32	33	34	35	36				
2	300	40	0.5	10	3				

HINTS

- Height of water which exerts same pressure as oil is : $h\rho_w g = 5\rho_{oil} \times g \Rightarrow h = 5 \times 0.8 = 4m$
Total height = $10 + 4 = 14m \Rightarrow v = \sqrt{2 \times 10 \times 14} = \sqrt{280} m/s$.
- Let mercury level drops in the left side by x mercury on right side would rise by $\frac{x}{n^2}$
Equating pressure at A & B

$$h\rho g = \left(x + \frac{x}{n^2}\right)\rho s g \Rightarrow \frac{x}{n^2} = \frac{h}{(n^2+1)s}$$



3. Initially, $w - 2kx = 0 \dots(i)$ Finally, $w' - 2k\left(x + \frac{a}{2}\right) - \frac{a^3}{2} \cdot 2\sigma g = 0 \dots(ii)$

$$w' = w + w_0 \Rightarrow w_0 = a(k + a^2\sigma g)$$

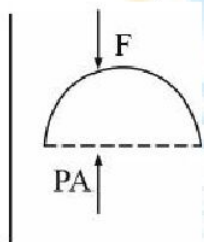
4. $Mg + 2T_1 \cos 45^\circ = F_b = \left(\frac{4}{3}\pi R^3\right)\rho_w g \therefore T_1 = \left(\frac{\frac{4}{3}\pi R^3\rho_w g - Mg}{\sqrt{2}}\right)$

5. Loss is $GPE = \frac{m}{\rho'}\rho g h_1 \Rightarrow h_1 = \frac{\rho'h}{\rho - \rho'} \Rightarrow mg(h + h_1) = \frac{m}{\rho'}\rho g h_1 \Rightarrow h_1 = \frac{\rho'H}{\rho - \rho'}$

6. $PA - F = F_b = \frac{2\pi}{3}r^3\rho_1 g$

$$(P_0 + \rho_1 g h)\pi r^2 - F = \frac{2\pi}{3}r^3\rho_1 g$$

$$F = P_0\pi r^2 + \left(h + \frac{2}{3}r\right)\pi r^2\rho_1 g$$



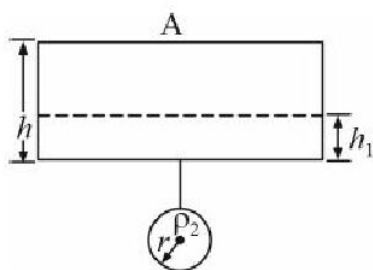
7. Taking cylinder and the ball as system

$$\frac{4}{3}\pi r^3 \cdot \rho_2 \cdot g + Ah \cdot \rho_1 g = \frac{4}{3}\pi r^3 \cdot \rho_w \cdot g + Ah_1 \cdot \rho_w g \Rightarrow r = \left[\frac{3A(h_1\rho_w - h\rho_1)}{4\pi(\rho_2 - \rho_w)}\right]^{1/3}$$

$$A = 11\text{cm}^2; h_1 = 4\text{cm}; \rho_w = 1\text{gm/cm}^3$$

$$\rho_1 = 0.5\text{gm/cm}^3; \rho_2 = 8\text{gm/cm}^3$$

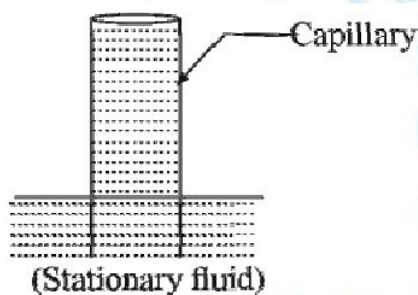
$$r = \left[\frac{3 \times 11(4 \times 1 - 6 \times 0.5)}{4 \times \left(\frac{22}{7}\right) \times (8 - 1)}\right]^{1/3} = \left(\frac{3}{8}\right)^{1/3} \text{cm}$$



8. Let the density of water be ρ , then the force by escaping liquid on container = $\rho S(\sqrt{2gh})^2$

$$\therefore \text{Acceleration of container } a = \frac{2\pi Sgh - \mu\rho vg}{\rho v} = \left(\frac{2Sh}{u} - \mu\right)g$$

Now $\mu = \frac{Sh}{v} \therefore a = \frac{Sh}{v}g$



9. $\frac{P_1}{\rho} + \frac{V_1^2}{2} = \frac{P_2}{\rho} + \frac{V_2^2}{2} \Rightarrow P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$

But, $P_1 - P_2 = \rho gh = \frac{\rho}{2}(V_2^2 - V_1^2) \Rightarrow V_2^2 - V_1^2 = 2gh$

10. $\tan \theta = \frac{h}{c} = \frac{a}{g} \Rightarrow a = \frac{hg}{c}$

Maximum volume that can be retained = $\frac{1}{2} \times h \times c \times b$

And $F = \left[M + \frac{hcb\rho}{2}\right] \frac{hg}{c}$

11. $T + B = m_1g$

$N = B + m_2g$

$\Rightarrow N = (m_1 + m)g - T$

12. Conceptual

13. $F_{\text{Buoyant}} = (m_A + m_B)g$; $2vd_{Fg} = v(d_A + d_B)g$

$d_A + d_B = 2d_F$. Therefore A, B, D

14. By equation of continuity,

$A_1V_1 = A_2V_2$

$$S_1 \times u = S_2 \times u \cos \theta \Rightarrow \theta = \cos^{-1} \left(\frac{S_1}{S_2} \right) \therefore \sin \theta = \sqrt{1 - \frac{S_1^2}{S_2^2}}$$

$$\text{Now range } R = \frac{u^2 \sin^2 2\theta}{g} = \frac{u^2 2 \sin \theta \cos \theta}{g} = \frac{2u^2}{g} \frac{S_1}{S_2} \sqrt{1 - \frac{S_1^2}{S_2^2}}$$

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{u^2}{2g} \left(1 - \frac{S_1^2}{S_2^2} \right)$$

$(S_1 \times u) \quad (S_2 \times u \cos \theta)$

$$15. \quad P_1 + \rho gh_1 + \frac{1}{2} \rho V_1^2 = P_2 + \rho gh_2 + \frac{1}{2} \rho V_2^2$$

$$h_1 = h_2 = 0 \quad (\text{Horizontal pipe})$$

$$\text{and } A_1 V_1 = A_2 V_2$$

$$4(4) = V_2$$

$$16 = V_2$$

$$2.80 \times 10^5 + \frac{1}{2} \rho (4)^2 = P_2 + \frac{1}{2} \rho (16)^2$$

$$2.80 \times 10^5 + \frac{1}{2} \rho (16 - 256) = P_2$$

$$2.80 \times 10^5 + \frac{1}{2} 900 (-240) = P_2$$

$$172 \times 10^3 \text{ N/m}^2 = P_2$$

$$F_1 = P_1 A_1 = 56 \times 10^3 ; F_2 = P_2 A_2 = 8.6 \times 10^3$$

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_{\text{pipe}} = \frac{d\vec{P}_{\text{oil}}}{dt} ; \vec{F}_{\text{pipe}} = \frac{d\vec{P}_{\text{oil}}}{dt} - (\vec{F}_1 + \vec{F}_2)$$

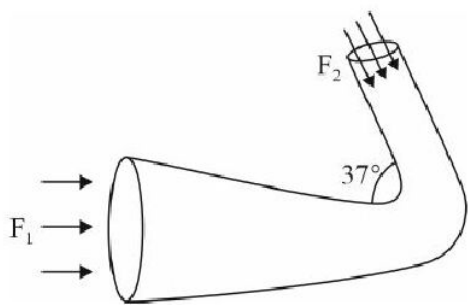
$$\vec{F} = P_1 A_1 \hat{i}$$

$$\vec{F}_2 = P_2 A_2 (\cos 37^\circ \hat{i} - \sin 37^\circ \hat{j}) ; \frac{d\vec{P}_{\text{oil}}}{dt} = \left(\frac{dm}{dt} \right) \Delta \vec{V} = (\rho A_1 V_1) (\vec{V}_2 - \vec{V}_1)$$

$$\vec{V}_2 = 16 [\cos 37^\circ (-\hat{i}) + \sin 37^\circ (\hat{j})]$$

$$\vec{V}_1 = 4\hat{i}$$

$$\text{Solving this for } F_{\text{pipe}} \text{ we get, } |F| = 76 \times 10^3 \text{ N}$$



16. $dA = 2\pi \times dy \times \frac{1}{\cos \theta}$ and

$$dF = \eta \frac{2\pi \times \sqrt{R^2 + h^2}}{h} \times \frac{V}{t} dy ; d\tau = \frac{\eta 2\pi}{ht} \times \sqrt{R^2 + h^2} \times \omega \cdot x^3 dy$$

$$\therefore \tan \theta \frac{R}{h} = \frac{x}{y} ; dy = \frac{h}{R} dx \Rightarrow \tau \frac{2\pi \eta \omega \sqrt{R^2 + h^2}}{Rt} \int_0^R x^3 dy, P = \tau \omega$$

17. Weight of cylinder (W) = $\pi(2r)^2 \times h \frac{\rho}{3} g = \pi \times 4r^2 \frac{h\rho}{3} .g$

Force due to pressure (P_1) created by liquid of height h_1 above the wooden block is

$$= P_1 \times \pi(2r)^2 = (P_0 + h_1 \rho g) \pi(2r)^2 = [P_0 + h_1 \rho g] \pi \times 4r^2$$

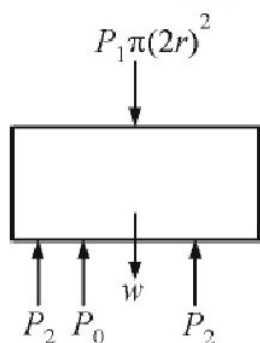
Upward direction due to pressure P_2 exerted from below the wooden block

$$\text{Block and atm pressure is } P_2 \pi [(2r)^2 - (r)^2] + P_0 \pi (r)^2$$

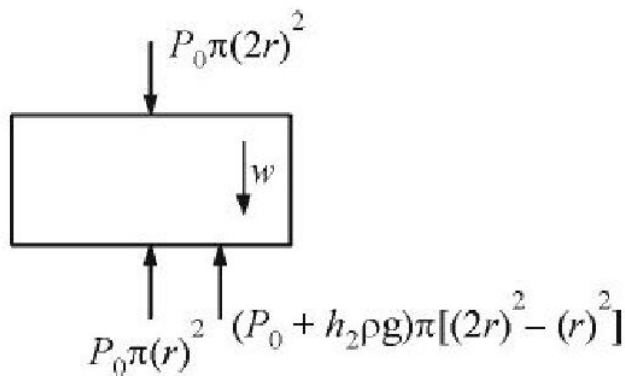
$$= [P_0 + (h_1 + h) \rho g] \pi \times 3r^2 + \pi r^2 P_0$$

At the verge of rising

$$[P_0 + (h_1 + h) \rho g] \pi \times 3r^2 + \pi r^2 P_0 = \pi \times 4r^2 h \times \frac{\rho}{3} g + [P_0 + h_1 \rho g] \times \pi \times 4r^2 \Rightarrow h_1 = \frac{5}{3} h$$



18. Balancing forces $h_2 = \frac{4h}{9}$



19. When the height h_2 of water level is further decreased then the upward force acting on the wooden block decreases. The total force downward remains the same. This difference will be compensated by the normal reaction by the wall on the wooden block. The block does not move up and remains at its original position.

20. Conceptual

21. $mg = F_b$

$$a^3 \times 0.4 \rho g = a^2 h \rho g \quad ; \quad h = \frac{2}{5} a = 0.4g$$

$$m = a^2 \times 0.4 \rho \quad ; \quad F_{net} = a^2 x \rho g$$

$$T = 2\pi \sqrt{\frac{a^3 \times 0.4 \rho}{a^2 \rho g}} \quad ; \quad T = 2\pi \sqrt{\frac{2a}{5g}}$$

22. Displacement must be less than submergence depth of cube.

23. $v_1 \rho_w g = (v - \Delta v) \rho_i g + \Delta v (\rho_m) g$

$$v_2 \rho_w g = (v - \Delta v) \rho_i g + \Delta v (\rho'_m) g$$

$$\frac{v_1}{v_2} = \frac{(v - \Delta v) \rho_i + \Delta v \rho_m}{(v - \Delta v) \rho_i + \Delta v \rho'_m} = \frac{\left(\frac{v}{\Delta v} - 1\right) \rho_i + \rho_m}{\left(\frac{v}{\Delta v} - 1\right) \rho_i + \rho'_m} = \frac{\left(\frac{1000}{20} - 1\right) \times 0.9 + 4.9}{\left(\frac{1000}{20} - 1\right) \times 0.9 + 1.9} = \frac{49}{46}$$

24. Mass of cavity is more in cube A than in cube B.

25. So long as cubes are floating, respective water levels do not change.

Let at $t = t_0$, cube A sinks.

$$v \rho_w g = (v - \Delta v) \rho_i g + \Delta v \rho_m g$$

v is volume of cube which is changing linearly with time at $t > t_0$.

$$v \rho_w g + N_A = (v - \Delta v) \rho_i g + \Delta v \rho_m g$$

After sinking water level decreases due to melting of ice,

$$\frac{1}{10} \frac{dv}{dt} = A \frac{dh_1}{dt}, A - \text{cross-section area of vessel}$$

Let at $t = t'_0$, cube B sinks

$$; \quad v' \rho_w g = (v' - \Delta v) \rho_i g + \Delta v \rho'_m g$$

$$\frac{1}{10} \frac{dv'}{dt} = A \frac{dh_2}{dt} \quad \therefore \frac{dv}{dt} = \frac{dv'}{dt} \Rightarrow \frac{dh_1}{dt} = \frac{dh_2}{dt}$$

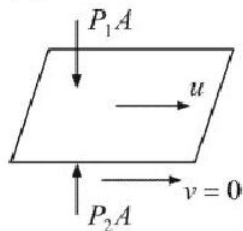
Final heights are same in both reach.

26. (P) $F = P_c \times A = \frac{h\rho g}{2} \times A$

(Q) $F_B = V\rho g = (hA)\rho g$

(R) $F_R = \frac{\Delta P}{\Delta t} = \frac{(V\Delta t)\rho u}{\Delta t} = \rho u^2 A$

(S)



$$P_1 + \frac{1}{2}\rho u^2 = P_2 + 0$$

$$(P_2 - P_1) = \frac{1}{2}\rho u^2$$

$$F = (P_2 - P_1)A = \frac{1}{2}\rho u^2 A$$

27. $\frac{F}{A} = \eta \frac{(\omega R)}{h}$; $F = \frac{(2\pi R l)\eta \omega R}{h}$; $\tau = F \times R$; Power = $\tau \omega$

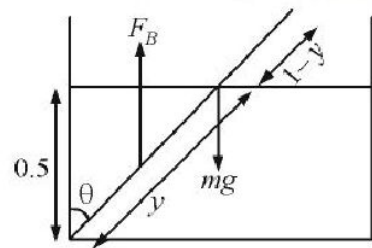
28. The force on small element of the slit $dF = \rho(dA)v^2$

$$dF = \rho(bdx) \times 2g(h-x) \Rightarrow F = 2\rho b g \int_0^l (h-x) dx = 2\rho b g \left(hl - \frac{l^2}{2} \right)$$

On putting values we get $F = 5N$

29. $F_B = (yA)\rho_w g$

$$mg = 1 \times A \times \frac{\rho_w}{2} \times g$$



Balancing torque $yA\rho_w g \times \frac{y}{2} \sin \theta = 1 \times A \times \frac{\rho_w}{2} \times g \times \frac{1}{2} \cos \theta \Rightarrow \cos^2 \theta = \frac{1}{2} \Rightarrow \frac{1}{\cos^2 \theta} = 2$

30. $\left(1 - \frac{1}{3}\right) V d_w = \left(1 - \frac{5}{9}\right) V d_l$

Therefore, $d_l = \frac{2}{3} \times \frac{9}{4} d_w = 1.5 \times 1000 = 1500 \text{ kg/m}^3$

31. Applying Bernoulli's equation at cross - section 1 and 2.

$$P_{atm} + \rho gh + 0 = P_2 + 0 + 0$$

$$\Rightarrow P_2 = P_{atm} + \rho gh \dots\dots(i)$$

Again applying Bernoulli's equation at section 2 and 3

$$P_2 + 0 + 2\rho g \times \frac{h}{2} = P_{atm} + \frac{1}{2} 2\rho V^2 \dots\dots(ii)$$

$$\Rightarrow V = \sqrt{2gh} \dots\dots(iii)$$

This is required velocity of efflux

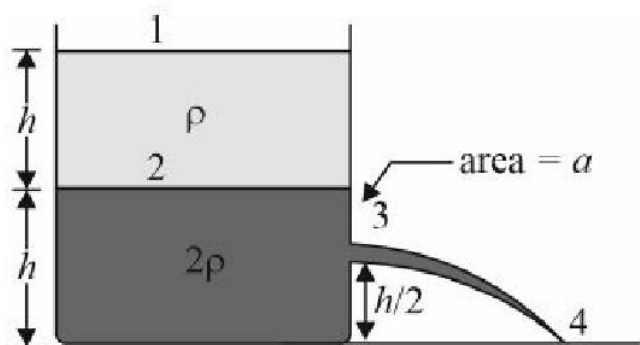
Applying continuity equation between 3 and 4 cross - section,

$$aV = a_1 V_1$$

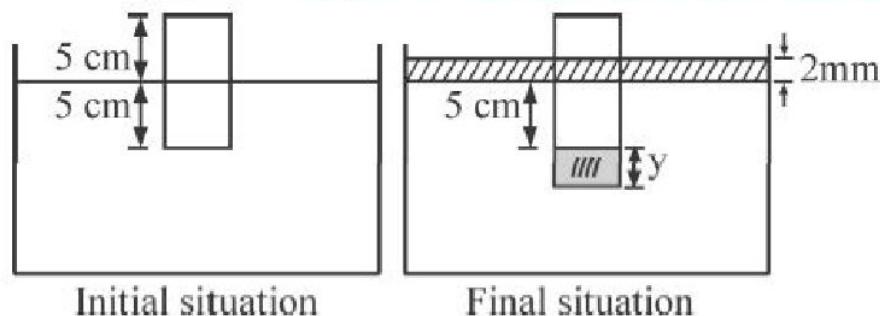
Again applying Bernoulli's equation between (iii) and (iv)

$$P_{atm} + \frac{1}{2} (2\rho) V^2 + 2\rho g \times \frac{h}{2} = P_{atm} + \frac{1}{2} (2\rho) V_1^2 + 0 \Rightarrow V_1^2 = 3gh$$

$$a_1 = \frac{aV}{V_1} = \frac{\sqrt{6} \cdot \sqrt{2gh}}{\sqrt{3gh}} = 2\text{cm}^2$$



32. Let the cube dips further by y cm and water level rises by 2 mm.



Then equating the volume (\\ \\ \\ volume = \\ \\ \\ volume in figure)

$\Rightarrow$ Volume of water raised = volume of extra depth of wood

$$\Rightarrow 100y = (1500 - 100) \frac{2}{10} = 1400 \times \frac{2}{10} = 280 \therefore y = 2.8\text{cm}$$

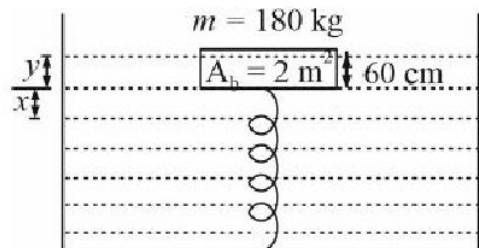
$\therefore$ Extra upthrust

$$\rho_{\text{water}} \times (2.8 + 0.2) \times 100g = mg$$

$$\Rightarrow m = 300 \text{ gm}$$

33. As block goes down by distance x , water comes up by distance y . As both are measured from initial level of water, compression in the spring is x but the block is in depth $(x + y)$ in water.

conservation of volume



$$0.2 \times x = (1\text{m}^2 - 0.2\text{m}^2) \cdot y$$

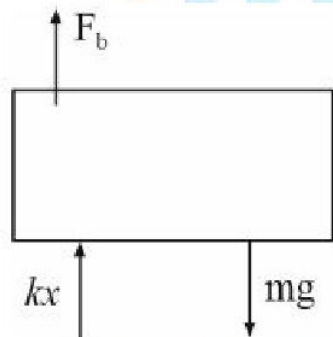
$$x = 4y$$

$$\Rightarrow y = \frac{x}{4}$$

$$\text{Thus total depth of block in water} = \frac{x}{4} + x = \frac{5x}{4}$$

Free body diagram in equilibrium:

$$F_b = (0.2) \left(\frac{5x}{4} \right) (1000)(10)$$



For equilibrium :

$$mg = kx + f_B$$

$$1800 = 2000x + (0.2) \left(\frac{5x}{4} \right) (1000)(10)$$

$$\Rightarrow 18 = 20x + 25x \Rightarrow x = \frac{18}{45} \text{ m} = 40 \text{ cm}$$

34. Conceptual

35. $R = 2\text{ cm}$

$$\Delta P = \frac{4S}{r} = \frac{4 \times 0.05}{2 \times 10^{-2}} = 10 \text{ Pa}$$

36. Let v be the velocity of the movable plate and F is equal to viscous force

$$F = \left[\eta_1 \frac{v}{h_1} + \eta_2 \frac{v}{h-h_1} \right] A \Rightarrow \frac{dF}{dh_1} = 0 \therefore h_1 = \frac{h}{3}$$

PROPERTIES OF MATTER

SINGLE ANSWER

1. Ice at 0°C is to be added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and content is 40°C . When a further 80 g of ice has been added and all it melted, the temperature of the whole content is 10°C . Calculate the specific latent heat of fusion of ice. [take $S_{\text{of water}} = 1 \text{ cal / gm } ^\circ\text{C}$.]

A) $3.8 \times 10^5 \text{ J / kg}$ B) $1.2 \times 10^5 \text{ J / kg}$ C) $2.4 \times 10^5 \text{ J / kg}$ d) $3.0 \times 10^5 \text{ J / kg}$

2. The specific heat of a metal at low temperature varies according to $S = aT^3$ where a is a constant and T is the absolute temperature. The heat energy needed to raise temperature of unit mass of the metal from $T = 1 \text{ K}$ to $T = 2 \text{ K}$ is:

A) $3a$ B) $\frac{15a}{4}$ C) $\frac{2a}{3}$ D) $\frac{12a}{5}$

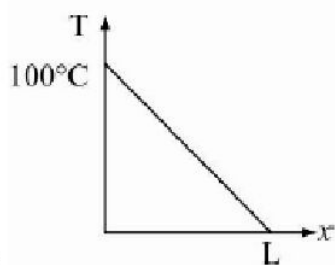
3. A thin copper wire of length L increase in length by 1% when heated from temperature T_1 to T_2 . What is the percentage change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2 ?

A) 1% B) 2% C) 3% D) 4%

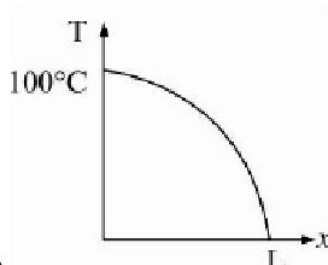
4. A cylinder of radius R made of a material of thermal conductivity k_1 is surrounded by a cylindrical shell of inner radius R and outer radius $2R$ made of a material of thermal conductivity k_2 . The two ends of the combined system are maintained at different temperatures. There is no loss of heat from the cylindrical surface and the system is in steady state. The effective thermal conductivity of the system is :

A) $k_1 + k_2$ B) $\frac{k_1 k_2}{k_1 + k_2}$ C) $\frac{1}{4}(k_1 + 3k_2)$ D) $\frac{1}{4}(3k_1 + k_2)$

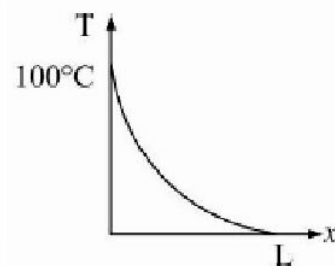
5. A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from $2K$ at end A to K at the other end B . The end A and B of the rod are maintained at constant temperature 100°C and 0°C , respectively. At steady state, the graph of temperature : $T = T(x)$ where x distance from end A will be



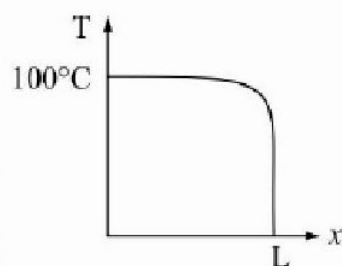
A)



B)



C)



D)

6. Two metallic spheres S_1 and S_2 are made of the same material and have got identical surface finish. The mass of S_1 is thrice that of S_2 . Both the spheres are heated to the same high temperature and placed in the same room having lower temperature but are thermally insulated from each other. The ratio of the initial rate of cooling of S_1 to that of S_2 is:

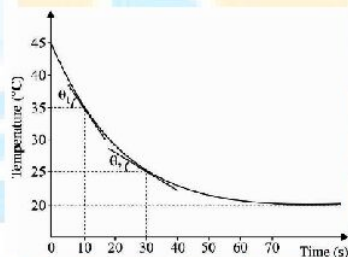
A) $\frac{1}{3}$

B) $\frac{1}{\sqrt{3}}$

C) $\frac{\sqrt{3}}{1}$

D) $\left(\frac{1}{3}\right)^{1/3}$

7. The temperature of a well stirred liquid kept open to a cold surrounding is plotted against time. The value of $\sec^2 \theta_1$ is (apply newton's law of cooling)



A) $1+9\tan^2 \theta_2$

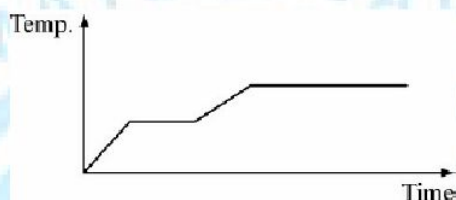
B) $1+\tan^2 \theta_2$

C) $1+3\tan^2 \theta_2$

D) $3\tan^2 \theta_2$

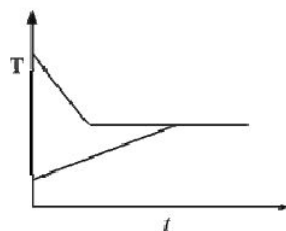
MULTIPLE

8. Four rods A, B, C, D of same length and material but of different radii $r, r\sqrt{2}, r\sqrt{3}$, and $2r$ respectively are held between two rigid walls. The temperature of all rods is increased by same amount. If the rods do not bend, then
- The stress in the rods are in the ratio $1 : 2 : 3 : 4$.
 - The force on the rod exerted by the wall are in the ratio $1 : 2 : 3 : 4$.
 - The energy stored in the rods due to elasticity are in the ratio $1 : 2 : 3 : 4$.
 - The strains produced in the rods are in the ratio $1 : 2 : 3 : 4$.
9. Heat is supplied to a certain homogenous sample of matter, at a uniform rate. Its temperature is plotted against time, as shown. Which of the following conclusions can be drawn?



- Its specific heat capacity is greater in the solid state than in the liquid state
 - Its specific heat capacity is great in the liquid state than in the solid state
 - Its latent heat of vaporization is greater than its latent heat of fusion
 - Its latent heat of vaporization is smaller than its latent heat of fusion
10. Two metallic spheres A and B are made of same material and have got identical surface finish. The mass of sphere A is four time that of B . Both the sphere are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other.
- The ratio of rate of heat loss of A to that of B is $2^{4/3}$.
 - The ratio of rate of heat loss of A to that of B is $2^{2/3}$.
 - The ratio of the initial rate of cooling of A to that of B is $2^{-2/3}$.
 - The ratio of the initial rate of cooling of A to that of B is $2^{-4/3}$.
11. Iron floats over mercury because density of mercury is more than the density of iron. Coefficient of volumetric expansion of iron and mercury are $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and $180 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Now consider an iron cube floating in a bowl of mercury at 0°C . When temperature is increased upto 25°C , then choose the correct options.
- Cube is submerged more at 25°C
 - Cube is submerged less at 25°C
 - Percentage change in the fraction of volume submerged is about %
 - Percentage change in the fraction of volume submerged is about 0.4%

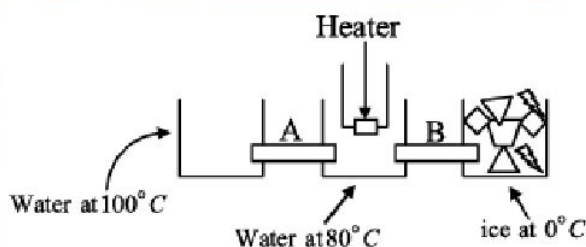
12. A sample A of liquid water and a sample B of ice of identical mass are kept in two neighbouring chambers in an otherwise insulated container. The chambers can exchange heat with each other. The graph of temperatures of the two chambers is plotted with time. $S_{ice} = \frac{S_{water}}{2}$.



- A) Finally the contents in sample A is water
 B) Equilibrium temperature is freezing point of water
 C) Ice melts partly
 D) Finally the contents in sample B is ice only

PARAGRAPH – 13 - 15

Figure shows three containers. The leftmost container contains water (heat capacity $= 5 \text{ cal}/^\circ\text{C}$), initially at temperature 100°C . The middle container contains water maintained at 80°C with the help of a heater. The container at the right hand side contain ice at 0°C . There are two heat conducting rods A and B both having thermal resistance equal to 10°C sec/cal . Discard any heat loss in the surrounding (Latent heat of fusion of ice $= 80 \text{ cal/gr}$)



13. Initially what is power of heater required to maintain the temperature of middle container at 80°C .
 A) 12 cal/sec B) 6 cal/sec C) 18 cal/sec D) 20 cal/sec
14. Find the rate with which ice melts in the rightmost container.
 A) 0.1 g/sec B) 1 g/sec C) 10 g/sec D) 8 g/sec
15. The temperature of water in the leftmost container decreases with time. The rate with which the ice melts in the rightmost container :
 A) will decrease with time B) will increase with time
 C) will remain steady D) will increase and then decrease

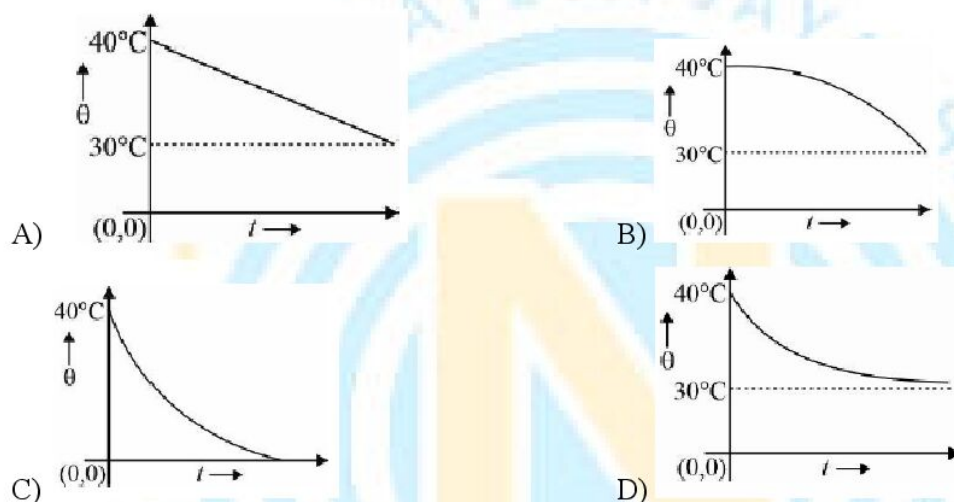
PARAGRAPH – 16 – 18

A body cools in a surrounding of constant temperature 30°C . Its heat capacity is $2\text{ J}/^\circ\text{C}$. Initial temperature of the body is 40°C . Assume Newton's law of cooling is valid. The body cools to 36°C in 10 minutes.

16. In further 10 minutes it will cool from 36°C to :

A) 34.8°C B) 32.1°C C) 32.8°C D) 33.6°C

17. The temperature of body in $^\circ\text{C}$ denoted by θ versus time t is best denoted as:

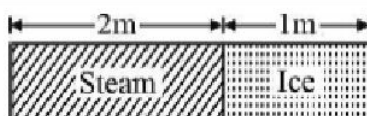
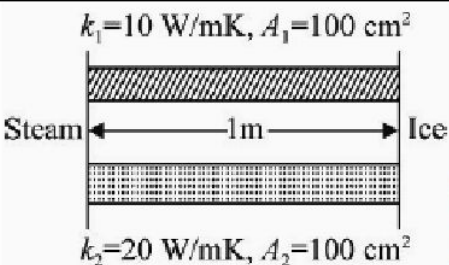
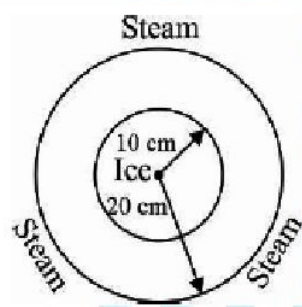
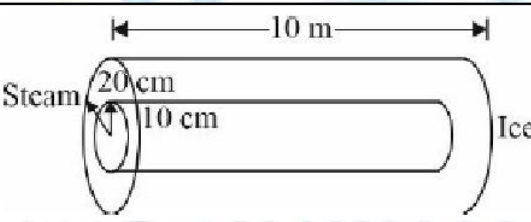


18. When the body temperature has reached 36°C , it is heated again so that it reaches to 40°C in 10 minutes. Assume that the rate of loss of heat at 38°C is the average rate of loss for the given time. The total heat required from a heater by the body is :

A) 7.52 J B) 0.728 J C) 16 J D) 32 J

MATRIX MATCHING

19. In column – I, certain situations are depicted where steam at 100°C is used to melt ice at 0°C by means of a conducting body which is insulated to prevent heat losses to surrounding. Match the numerical value of question asked in each entry to the corresponding entry in column – II. Symbols have usual meaning. All the cases of column – I are to be taken at steady state.

Column – I		Column – II	
A)	 $k_1 = 10 \text{ W/mK}$ $k_2 = 20 \text{ W/mK}$ Find temperature of interface of two rods in $^{\circ}\text{C}$.	p)	10
B)	 $k_1 = 10 \text{ W/mK}$, $A_1 = 100 \text{ cm}^2$ $k_2 = 20 \text{ W/mK}$, $A_2 = 100 \text{ cm}^2$ Find total rate of heat transfer through the two rods in SI units	q)	20
C)	 Hollow sphere of $k = \frac{1}{32\pi} \text{ W/mK}$ Find thermal resistance in SI units	r)	30
D)	 Hollow cylinder of $k = \frac{\pi}{10} \text{ W/mK}$ Find temperature gradients in SI units.	s)	40

NUMERICAL

20. 2 kg of ice at -15°C is mixed with 2.5 kg of water at 25°C in an insulating container. If the specific heat capacities of ice and water are $0.5\text{ cal/g}^\circ\text{C}$ and $1\text{ cal/g}^\circ\text{C}$ find the amount of water present in the container (in kg) ? (approximate it to nearest integer)
21. In a container of negligible heat capacity, 200 gm ice at 0°C and $\frac{262}{3}\text{ gm}$ steam at 100°C are added to 200 gm of water that has temperature 55°C . Assume no heat is lost to the surroundings and the pressure in the container is constant 1.0 atm . ($L_f = 80\text{ cal/gm}$, $L_v = 540\text{ cal/gm}$, $s_w = 1\text{ cal/gm}^\circ\text{C}$). Find the amount (in gm) of the steam left in the system.

PROPERTIES OF MATTER

KEY

1	2	3	4	5	6	7	8	9	10
A	B	B	C	B	D	A	BCD	BC	AC
11	12	13	14	15	16	17	18	19	20
AD	BD	B	A	C	D	D	C	A-q; B-r; C-s; D-p	3
21									
4									

HINTS

1. Let x be latent heat of fusion w be water equivalent of apparatus

$$200(70 - 40) + w(70 - 40) = 50(x + 40) \dots (i)$$

$$250(40 - 10) + w(40 - 10) = 80(x + 10) \dots (ii)$$

$$\text{Solving (i) and (ii) } x = 90\text{ cal/g} \approx 3.8 \times 10^5\text{ J/kg}$$

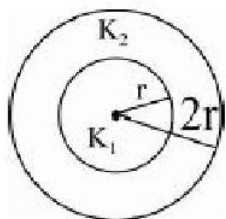
$$2. \quad dq = msdT \Rightarrow q = \int_1^2 1 \times aT^3 dT = \frac{15a}{4}$$

$$3. \quad \frac{dl}{l} = 1\% \quad \therefore \frac{dA}{A} = 2 \times 1\% = 2\%$$

$$4. \quad R_1 = \frac{l}{K_1 \pi r^2} \& R_2 = \frac{l}{3K_2 \pi r^2}$$

Both rods are in parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{4\pi r^2 K_{eq}}{l} = \frac{\pi r^2 K_1}{l} + \frac{3\pi r^2 K_2}{l} \therefore K_{eq} = \frac{K_1 + 3K_2}{4}$$



5. $K(x) = 2K - \frac{Kx}{l}$

$$H = -K(x)A \left(\frac{dT}{dx} \right)$$

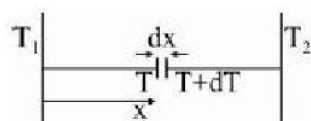
$$H = - \left[2K - \frac{Kx}{l} \right] A \left(\frac{dT}{dx} \right)$$

Integrate and get temperature as a function of x

Alternative approach

Initially K is higher $\therefore \frac{dT}{dx}$ should be smaller in magnitude

Later K decrease $\frac{dT}{dx}$ increases



6. $ms \frac{d\theta}{dt} = \sigma A (T^4 - T_0^4) \Rightarrow m_1 = 3m_2 \Rightarrow r_1 = 3^{1/3} r^2$

7. According to Newton's law of cooling $\frac{dT}{dt} \propto \Delta T$

$$\Rightarrow \frac{\tan \theta_1}{\tan \theta_2} = \frac{35 - 20}{25 - 20} = \frac{15}{5} = \frac{3}{1} \Rightarrow \tan^2 \theta_1 = 9 \tan^2 \theta_2$$

$$\therefore \sec^2 \theta_1 + 1 + \tan^2 \theta_1 \Rightarrow \sec^2 \theta_1 = 1 + 9 \tan^2 \theta_2$$

8. Thermal force $= Y A \alpha d\theta = Y \pi r^2 \alpha d\theta$

$$r_1 = r, r_2 = r\sqrt{2}, r_3 = r\sqrt{3}, r_4 = 2r \Rightarrow F_1 : F_2 : F_3 : F_4 = 1 : 2 : 3 : 4$$

Thermal stress $= Y \alpha d\theta$

As Y and α are same for all the rods, hence stress developed in each rod will be the same. As strain

$$= \alpha d\theta, \text{ so strain will also be the same. } E = \text{Energy stored} = \frac{1}{2} Y (\text{strain})^2 \times A \times L$$

$$\therefore E_1 : E_2 : E_3 : E_4 = 1 : 2 : 3 : 4$$

So, option (B) and (C) are correct.

9. We can see the slope of second step (liquid heating) is less than that of first step (solid heating) hence specific heat of liquid is greater than that of solid. We can also see that the horizontal portion of first part is smaller than that of second part which implies that solid melting is taking less time than liquid vaporization hence (C) is also correct.

$$18. \quad m_A = 4m_B \Rightarrow S_A = (4)^{2/3} S_B$$

(mass) (surface area)

$$\frac{H_A}{H_B} = \frac{S_A}{S_B} \text{ also } \frac{dT}{dt} \propto \frac{S}{m}$$

11. $\gamma_{Hg} > \gamma_{Fe} \Rightarrow$ density of Hg decrease more on heating. Hence cube will be submerged more.

If fraction submerged is f

$$\rho_{Hg} V f g = \rho_{Fe} V g \Rightarrow f = \frac{\rho_{Fe}}{\rho_{Hg}}$$

$$\text{Change of fraction} = \frac{\frac{\rho_{Fe}}{\rho_{Hg}} - \frac{\rho_{Fe}}{\rho_{Hg}}}{\frac{\rho_{Fe}}{\rho_{Hg}}} = \frac{1 + \gamma_{Hg} \times 25}{1 + \gamma_{Fe} \times 25} - 1$$

$$= \frac{1 + 180 \times 10^{-6} \times 25}{1 + 35 \times 10^{-6} \times 25} - 1 = \frac{(180 - 35) \times 10^{-6} \times 25}{1} = 3.6 \times 10^{-3} = 0.36\% = 0.4\%$$

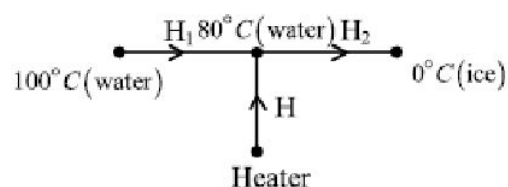
12. Conceptual

13-15

$$\text{Power of Heater } H = H_2 - H_1 = \frac{80 - 0}{10} - \frac{100 - 80}{10} = 6 \text{ cal / s}$$

$$H_2 = L_f \frac{dm}{dt} \left(\text{rate of melting} = \frac{dm}{dt} \right)$$

$$\Rightarrow \frac{80 - 0}{10} = 80 \frac{dm}{dt} \Rightarrow \frac{dm}{dt} = 0.1 \text{ gr / sec}$$



Since temperature of middle container is constant, H_2 is constant, hence the rate of melting of ice will be constant

$$16. \frac{40-36}{10} = k(36-30); \frac{39-x}{10} = k\left(\frac{36+x}{2}-30\right)$$

$$\frac{4}{36-x} = \frac{8}{\left(\frac{x}{2}-30+18\right)}; \frac{x}{2}-12 = 72-2x; \frac{5x}{2} = 72+12; \frac{5x}{2} = 84;$$

$$x = \frac{2 \times 84}{5} = \frac{168}{5} = 33.6$$

$$17. \frac{d\theta}{dt} = \frac{kA}{ms}(T-T_0) \text{ Magnitude of slope will decrease with time.}$$

$$18. \therefore \frac{40-36}{10} = k(38-30) \Rightarrow k = \frac{4}{10 \times 8} = \frac{1}{20}$$

When the block is at $38^\circ C$ and room temperature is at $30^\circ C$ the rate of heat loss

$$ms \times \frac{d\theta}{dt} = ms k (38-30)$$

$$19. (a) \frac{10}{2}(100-T) = \frac{20 \times A}{1}(T-0); 100-T-4T=0; T=20$$

$$(b) \frac{10 \times A(100)}{1} + \frac{20 \times A(100)}{1}; 3000 \times 100 \times 10^{-4} = 30W$$

$$(c) R_{th} = \frac{(0.2-0.1)}{4\pi \times 0.2 \times 0.1} = 40 \text{ SI units}$$

$$(d) \frac{dt}{dx} = \frac{100-0}{10} = 10^\circ C/m$$

20. Let $x kg$ of ice remaining

$$(2)(5) \times 500 + (2-x) \times 80,000 = (2.5)(25)(1000)$$

$$x = 1.4 kg \quad \therefore \text{Water} = 3 kg$$

$$21. Q_{ab} = 200 \times 80 + 200 \times 1 \times 100 + (200 \times 1 \times 45)$$

$$Q_{loss} = m_s (540)$$

$$16000 + 20000 + 9000 = (540)m_s; \frac{45000}{540} = m_s; m_s = \frac{250}{3} = 83.33$$

$$\text{Steam left } 87.33 - 83.33 = 4$$

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